



清华大学 交叉信息研究院
Institute for Interdisciplinary Information Sciences, Tsinghua University



Advanced Topics for Robotics 智能机器人前沿探究

Lecture 8: Robot Locomotion

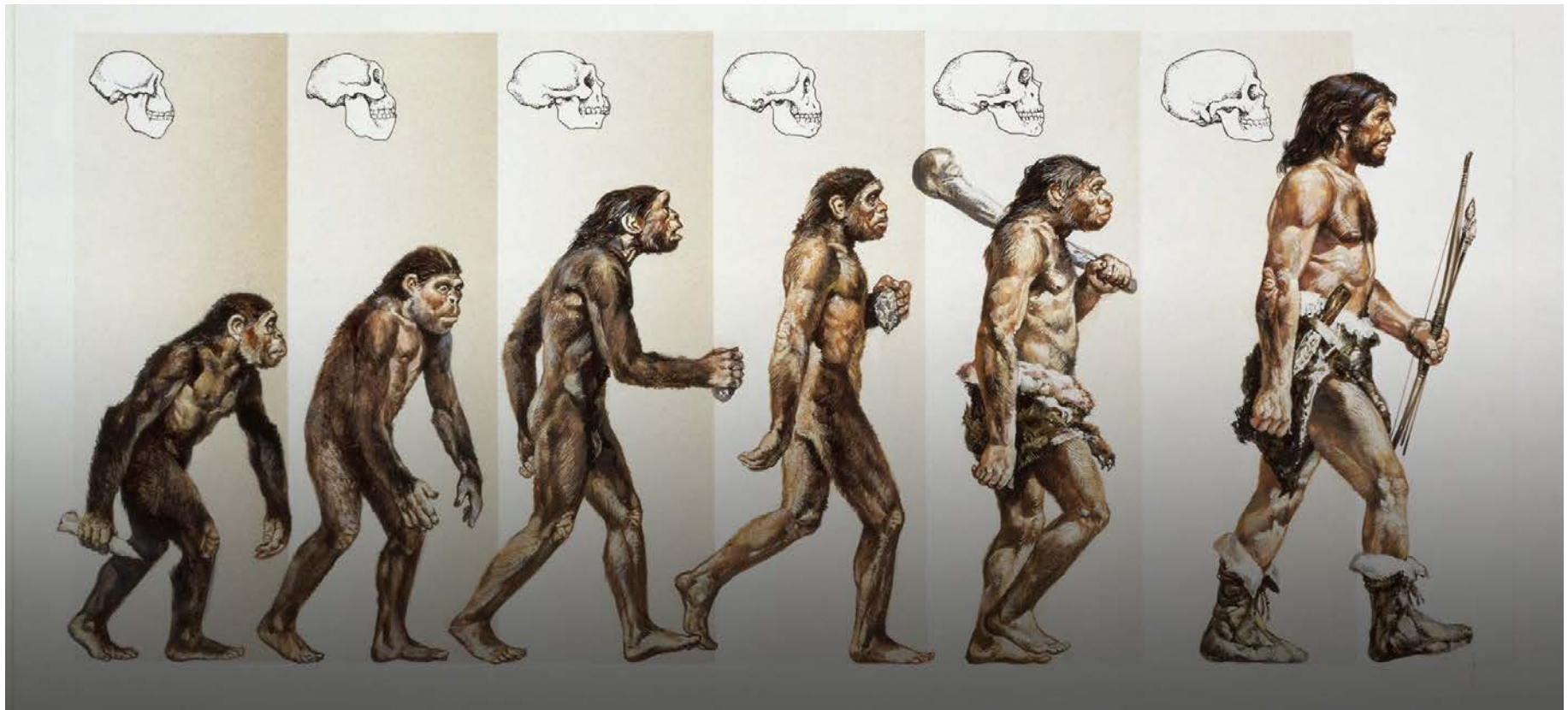
Jianyu Chen (陈建宇)

Institute for Interdisciplinary
Information Sciences

Tsinghua University

Legged locomotion for human

- Legged locomotion: The most important feature for human being



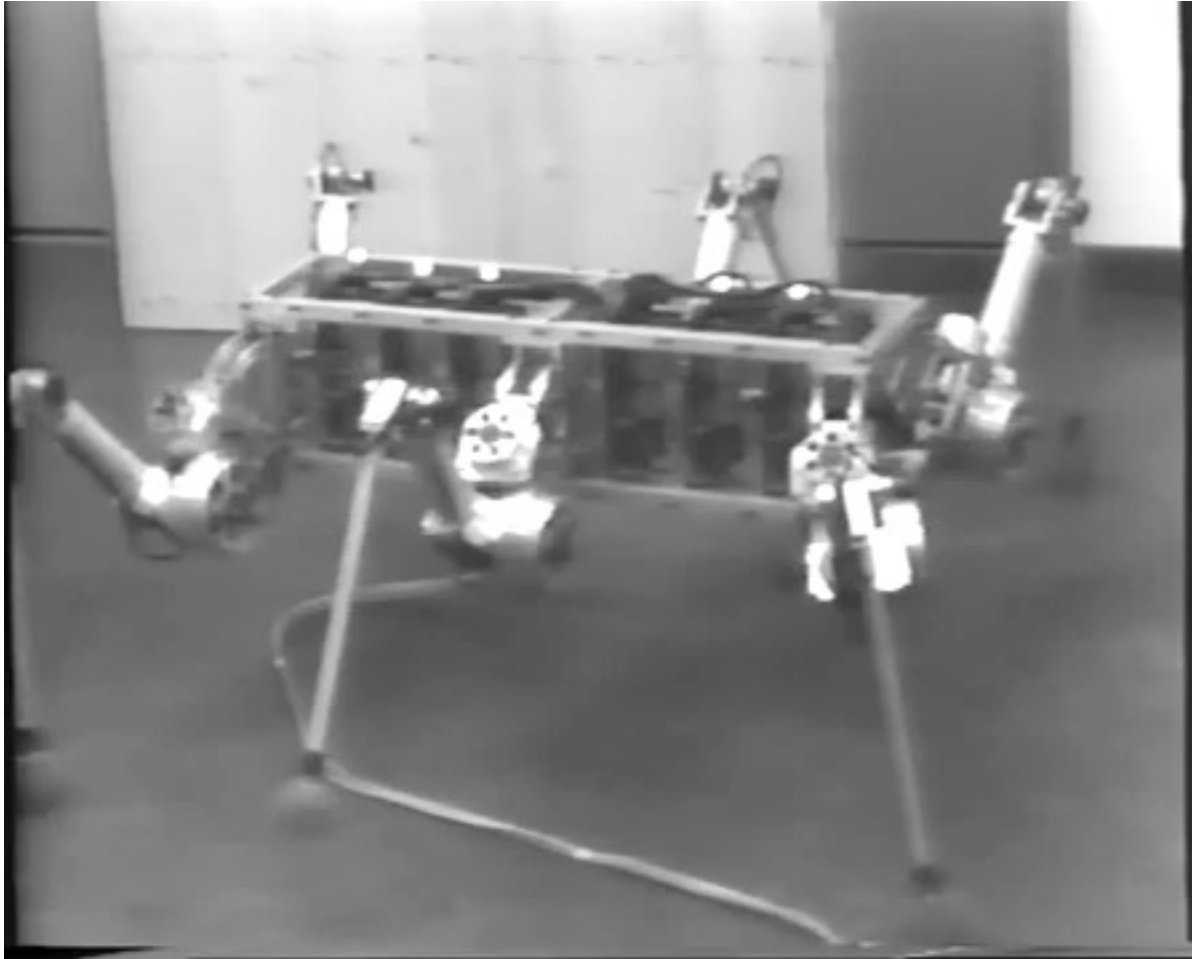
Historical legged locomotion machines

- ❑ People try to make legged locomotion machines through the history



The first walking six-legged robot

- ❑ Six-legged robot developed by CMU



Boston dynamics

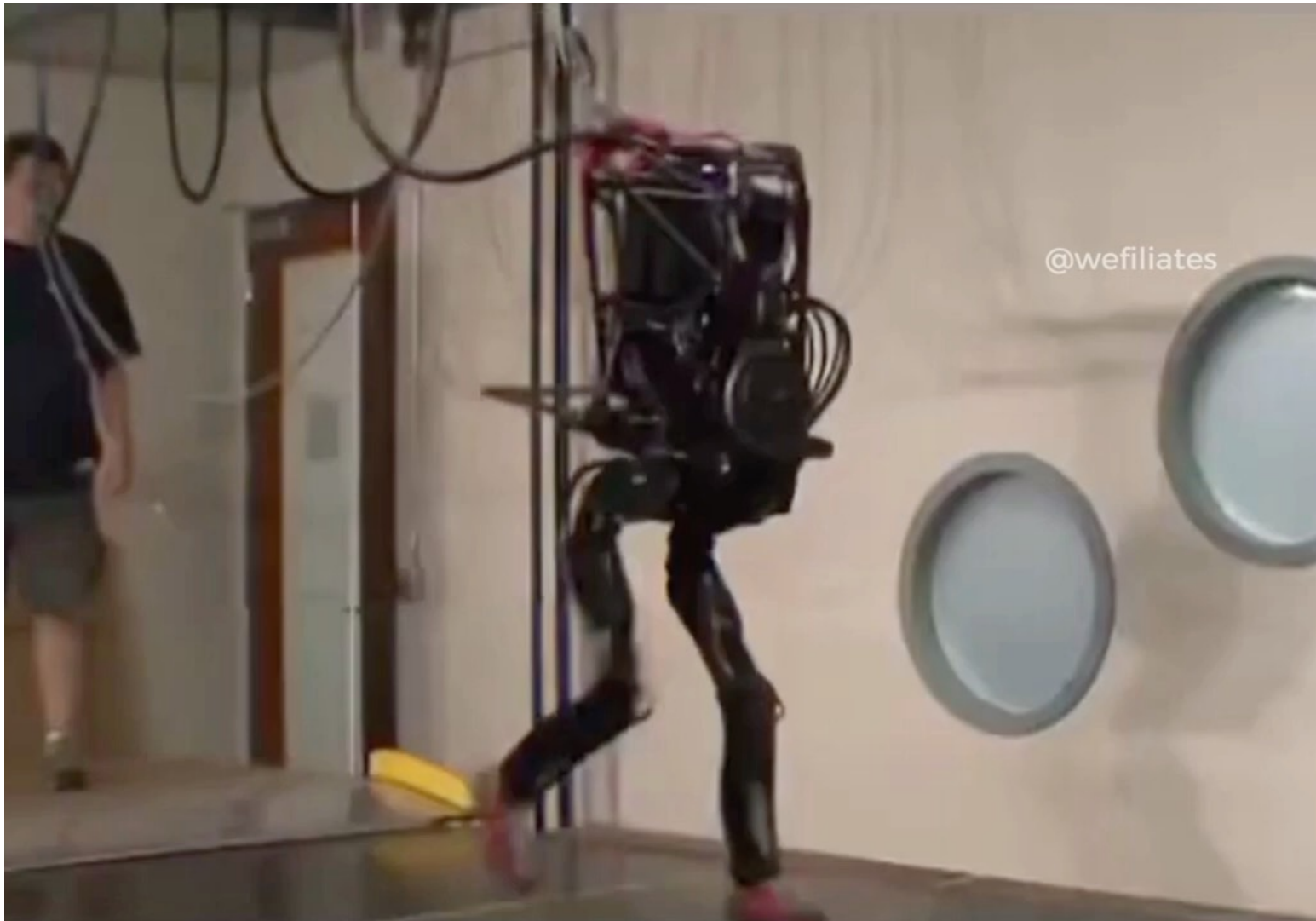
- ❑ Marc Raibert invented several legged robot in CMU and MIT, and then founded Boston Dynamics



M. Raibert (1980) (RI, CMU & AI lab, MIT): Hopping, Robots, Monoped, biped and Quadpeds.
Founded Boston Dynamics (1992)

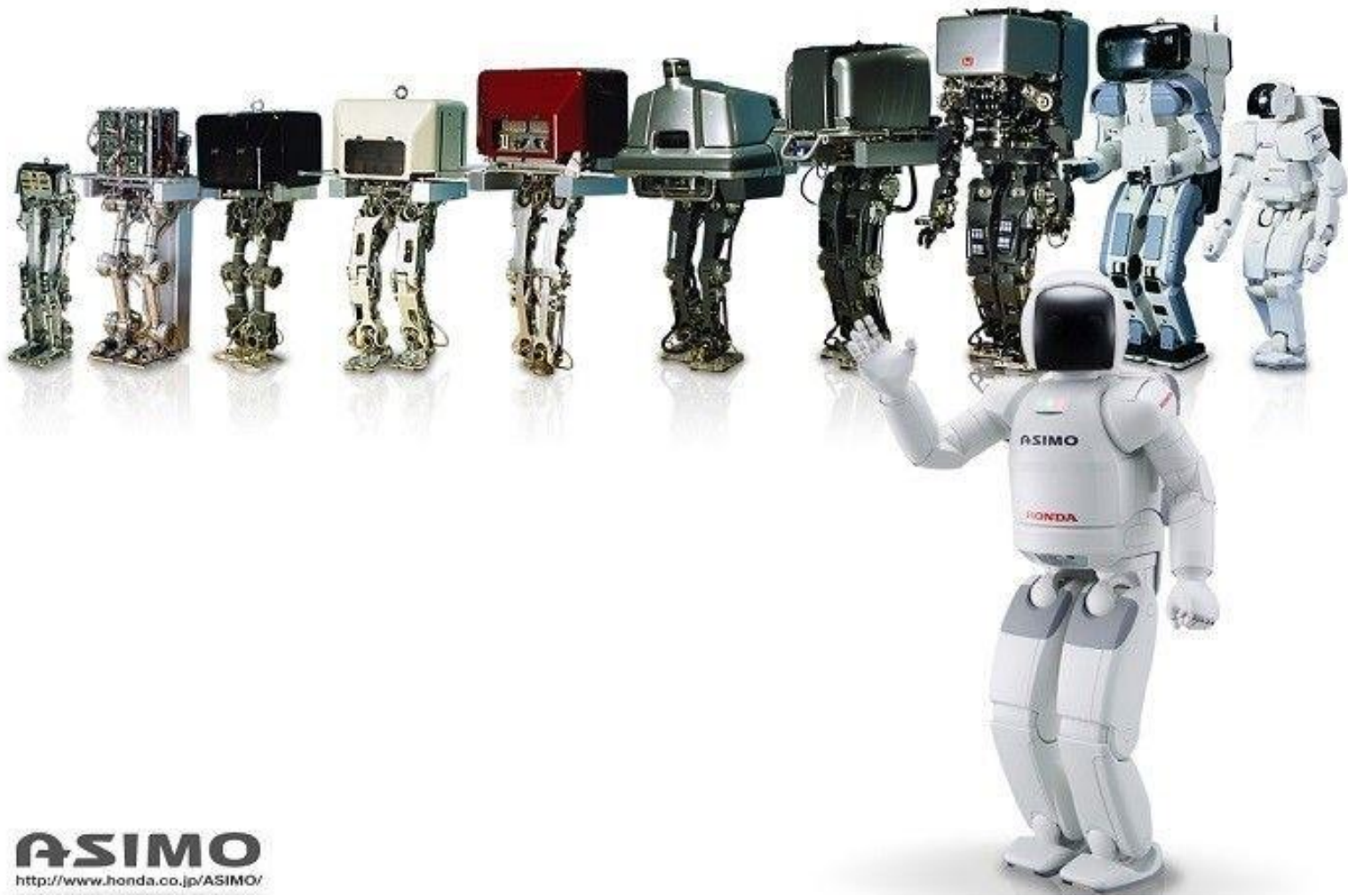
Boston dynamics

□ Evolution of Atlas robot



Honda ASIMO

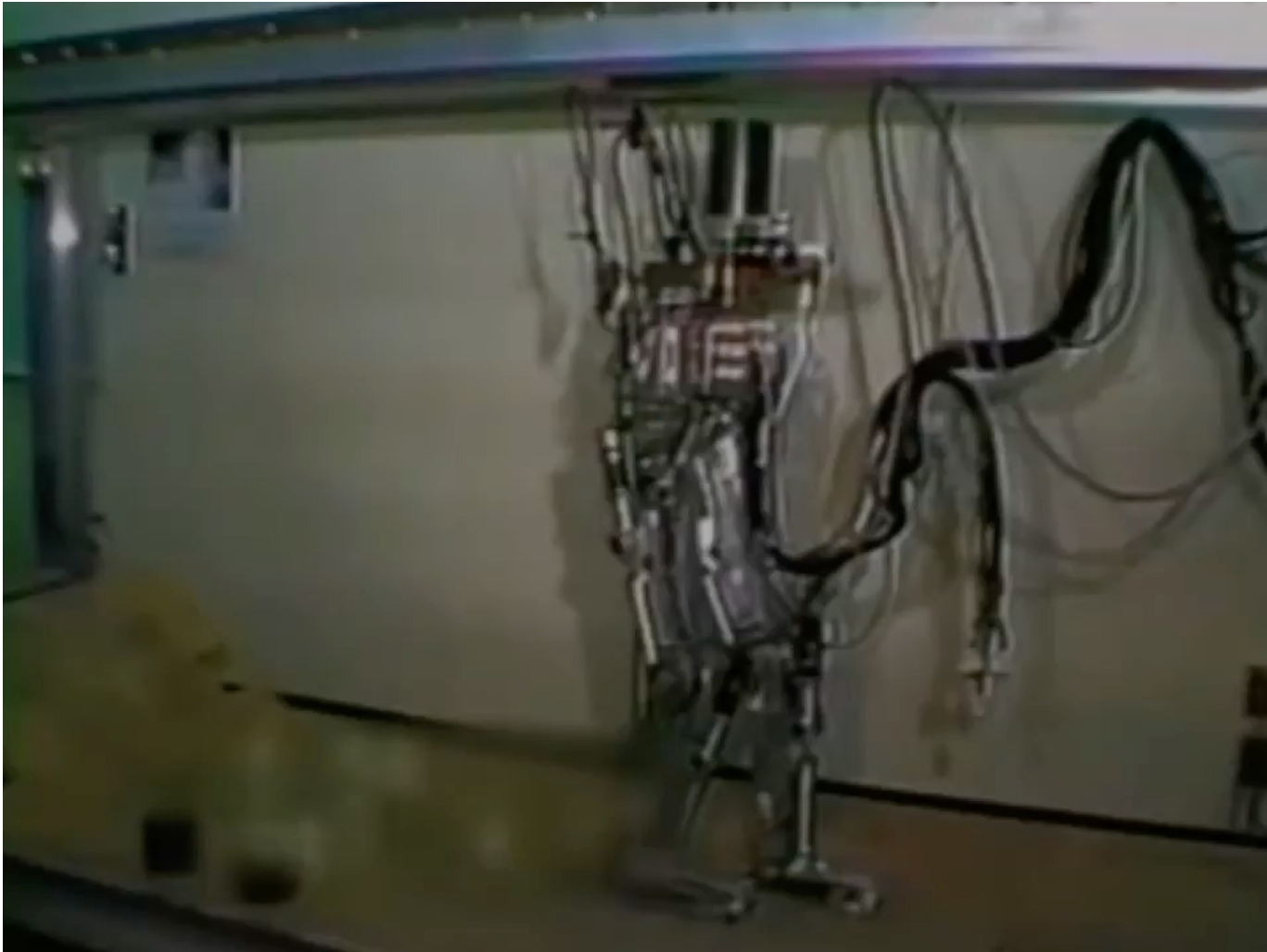
□ The evolution of ASIMO



ASIMO
<http://www.honda.co.jp/ASIMO/>

Honda ASIMO

□ The evolution of ASIMO



Jessie Grizzle and his students



Jessie Grizzle @UMich



Jonathan Hurst @OSU

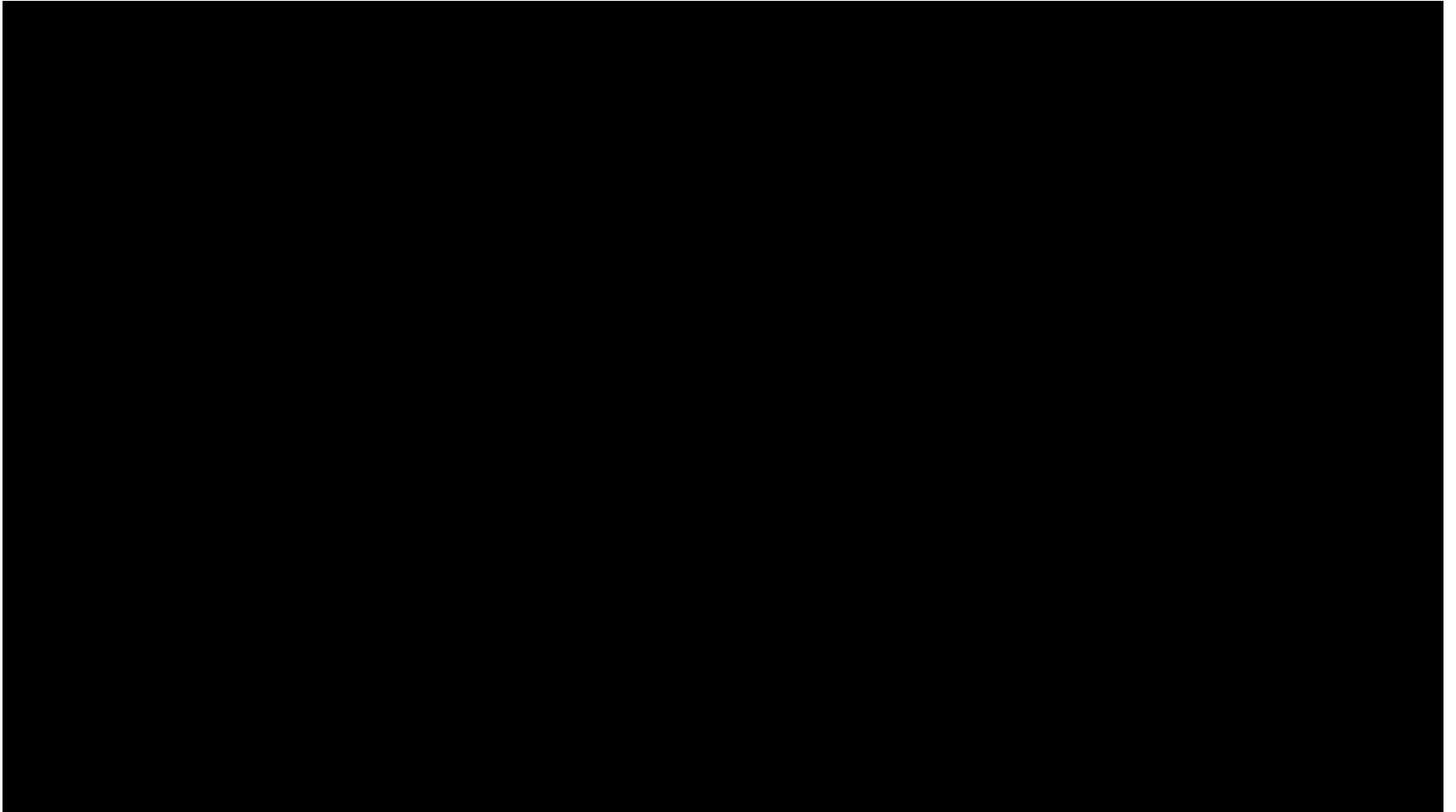


Aaron Ames @Caltech



Koushil Sreenath @Berkeley

Cassie robot



The MIT Cheetah



Sangbae Kim



Massachusetts Institute of Technology

Autonomous Running Jumps Over Obstacles in the MIT Cheetah 2

Hae-Won Park, Patrick Wensing, and Sangbae Kim



MITMECHE

BIOMIMETIC ROBOTICS LAB

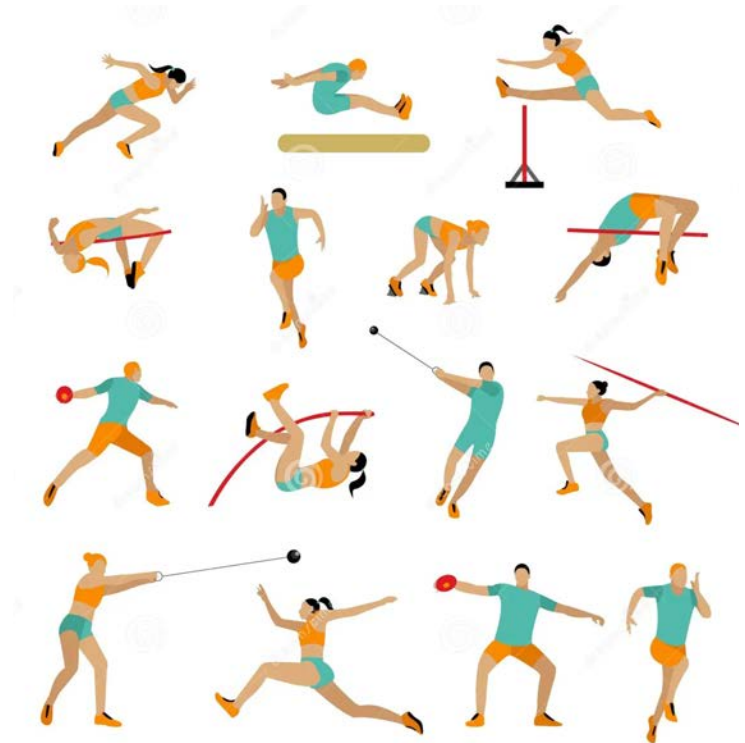


Marco Hutter

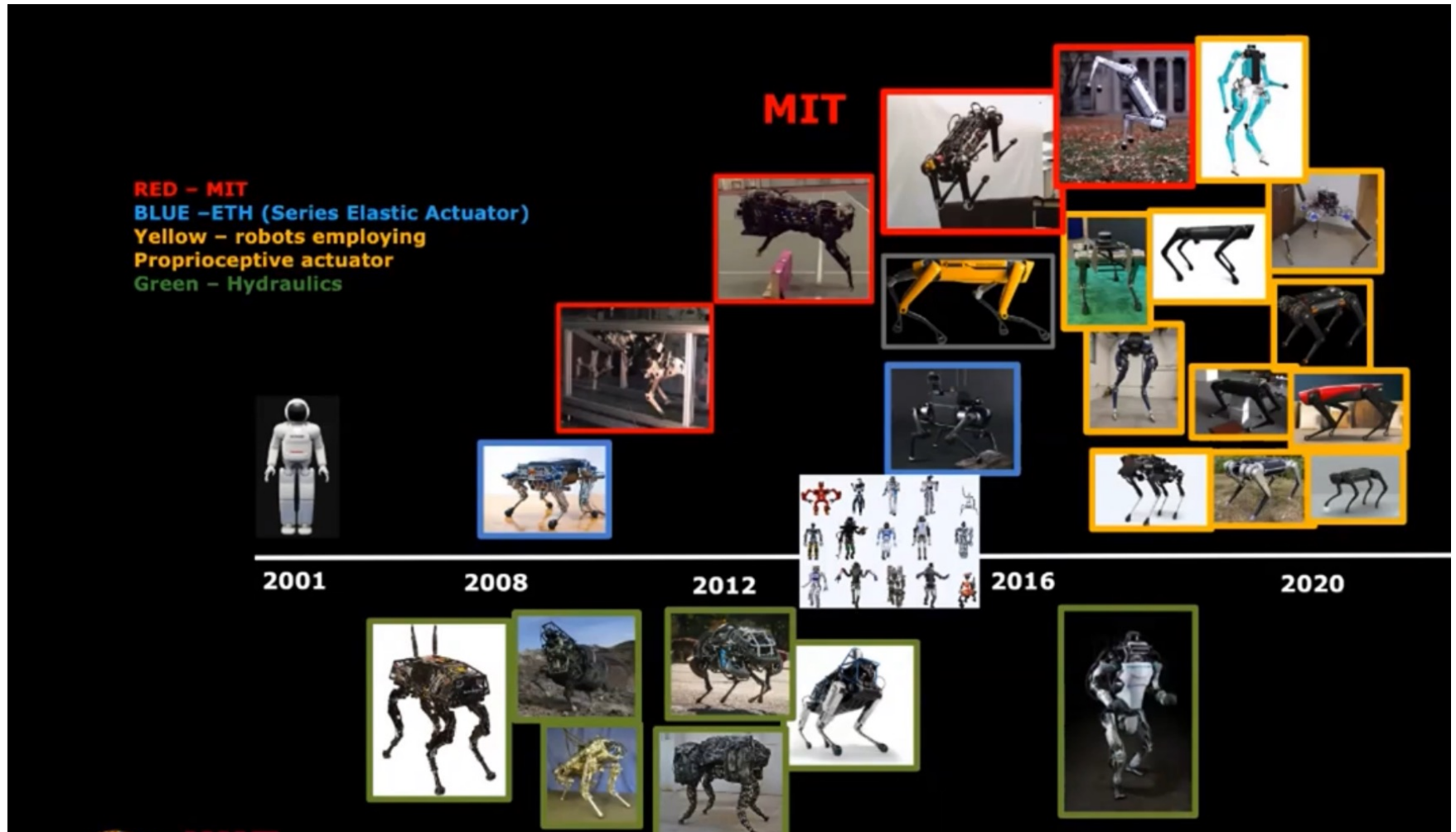


Goal for locomotion

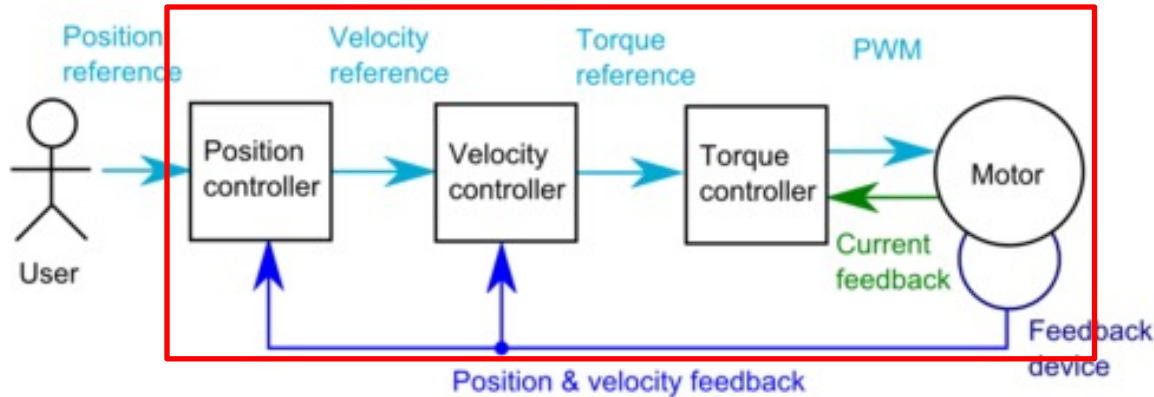
- ❑ What is the basics for locomotion?
- ❑ Control the robot to reach target state in the future
 - Body orientation, angular velocity
 - CoM position, velocity
 - Foot position, orientation
 - Contact state
 - Arm position, velocity
 - ...



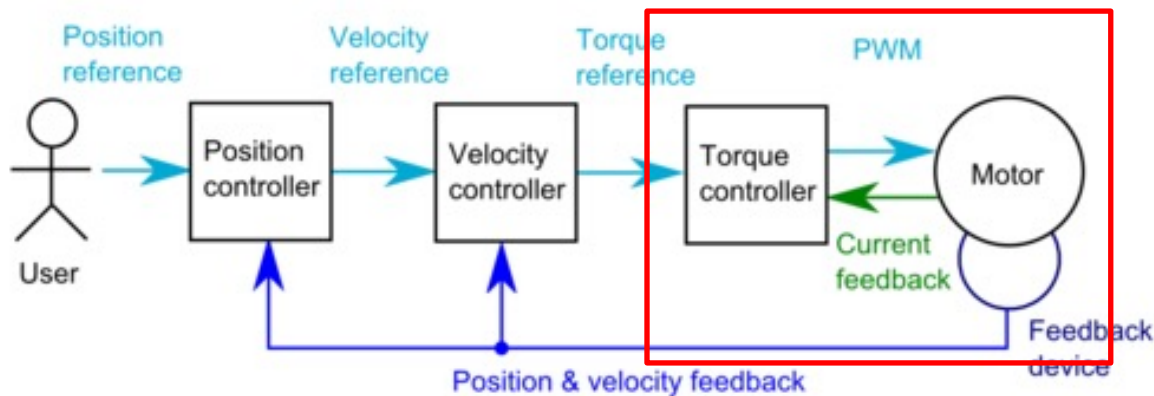
Types of actuation



Types of joint control



Position control



Torque control



Honda ASIMO

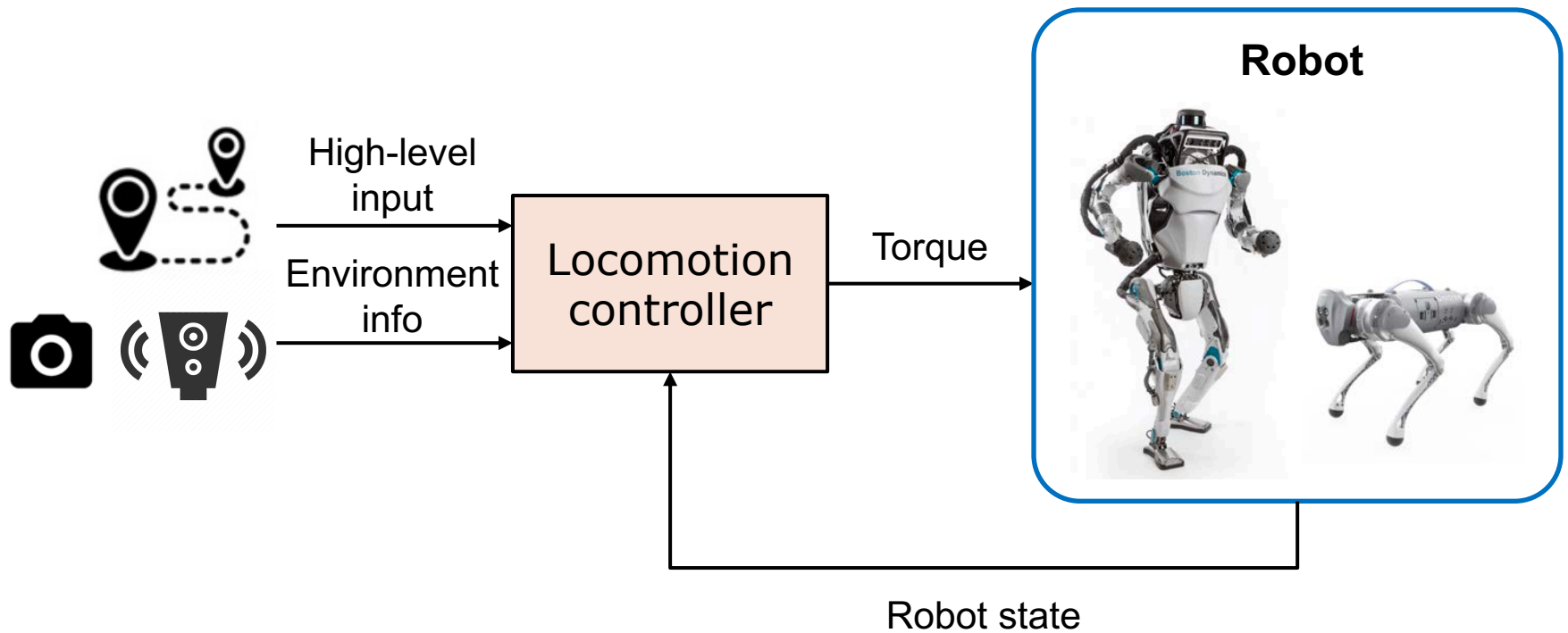
Best position-controlled robot
Project terminated at 2018



Boston Dynamics Atlas

Absolutely the best legged robot
Torque controlled

General problem formulation



General problem formulation

□ Inside the Locomotion Controller: Control as optimization formulation

$$u(\cdot), x(\cdot) = \operatorname{argmin}_{u(\cdot), x(\cdot)} l_T(x(T)) + \int_{t=0}^T l(x(t), u(t))$$

$$\text{s.t.} \left\{ \begin{array}{ll} M(q)\ddot{q} + C(q, \dot{q})\dot{q} = \tau_g(q) + Bu + J^T(q)\tau_c & \text{Dynamics constraint} \\ \phi(q) \geq 0 \quad \phi(q)\tau_c = 0 & \text{Contact constraint} \\ \tau_c \in \mathcal{C}(n, \mu) & \text{Friction constraint} \end{array} \right.$$

$$q = (q_B, q_J) \quad x = (q, \dot{q})$$

Challenges

□ Underactuation

$$u(\cdot), x(\cdot) = \underset{u(\cdot), x(\cdot)}{\operatorname{argmin}} l_T(x(T)) + \int_{t=0}^T l(x(t), u(t))$$

$$\text{s.t.} \left\{ \begin{array}{l} M(q)\ddot{q} + C(q, \dot{q})\dot{q} = \tau_g(q) + Bu + J^T(q)\tau_c \\ \phi(q) \geq 0 \quad \phi(q)\tau_c = 0 \\ \tau_c \in \mathcal{C}(n, \mu) \end{array} \right.$$

$$q = (q_B, q_J) \quad x = (q, \dot{q})$$

Only activated for joints



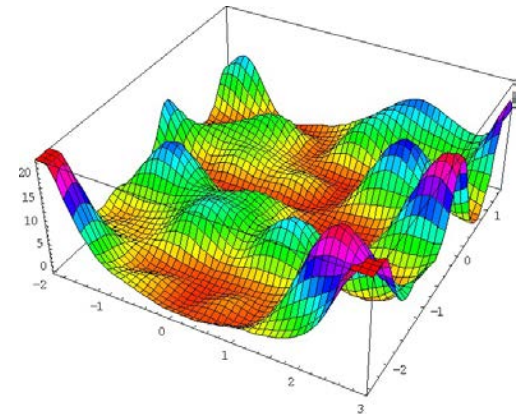
Challenges

□ High computation burden

$$u(\cdot), x(\cdot) = \operatorname{argmin}_{u(\cdot), x(\cdot)} l_T(x(T)) + \int_{t=0}^T l(x(t), u(t))$$

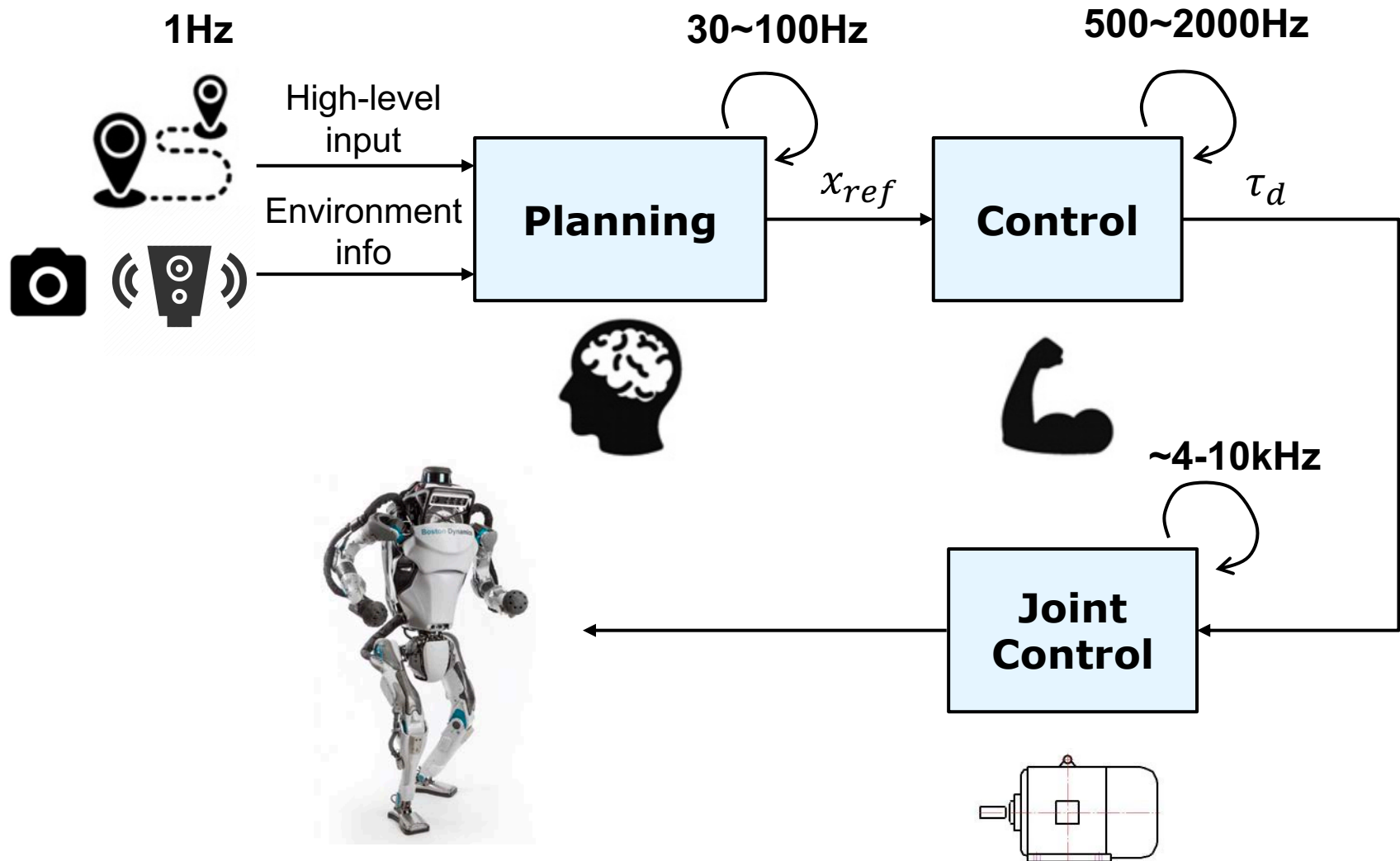
$$\text{s.t.} \left\{ \begin{array}{l} M(q)\ddot{q} + C(q, \dot{q})\dot{q} = \tau_g(q) + Bu + J^T(q)\tau_c \\ \phi(q) \geq 0 \quad \phi(q)\tau_c = 0 \\ \tau_c \in \mathcal{C}(n, \mu) \end{array} \right.$$

$$q = (q_B, q_J) \quad x = (q, \dot{q})$$



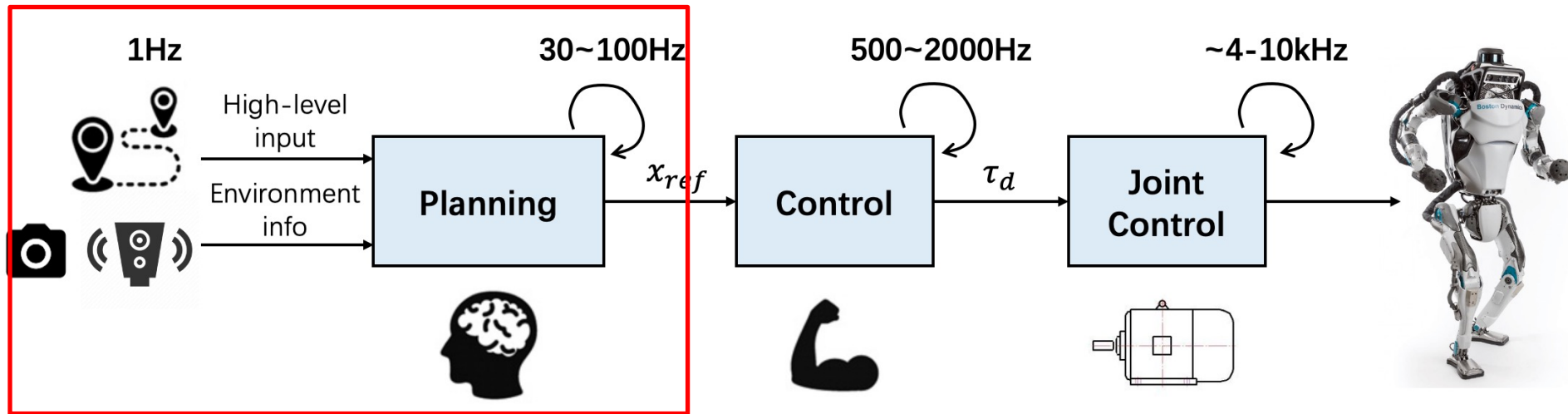
Hierarchical control framework

- Break down the complexity with **hierarchy**



Gait planning

□ Planning with **simplified dynamics model**



$$u(\cdot), x(\cdot) = \underset{u(\cdot), x(\cdot)}{\operatorname{argmin}} l_T(x(T)) + \int_{t=0}^T l(x(t), u(t))$$

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} = \tau_g(q) + Bu + J^T(q)\tau_c$$

s.t.

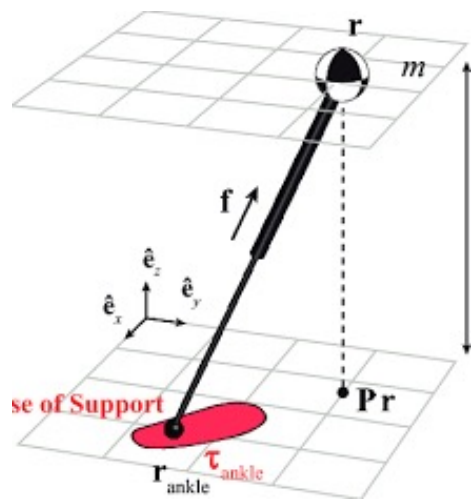
$$\phi(q) \geq 0 \quad \phi(q)\tau_c = 0$$

$$\tau_c \in \mathcal{C}(n, \mu)$$

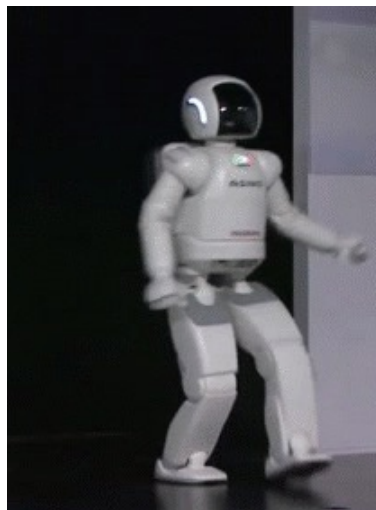
Simplify the model

Simplified dynamic models

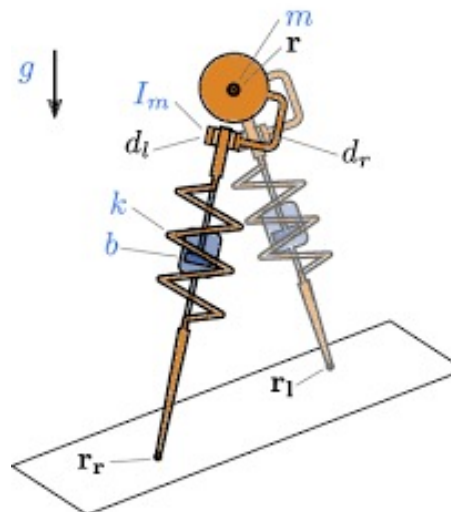
- ❑ Inverted pendulum models
- ❑ Largely simplified model **without** angular dynamics



Linear Inverted
Pendulum (LIP) model



ASIMO



Spring-loaded Inverted
Pendulum (SLIP) model

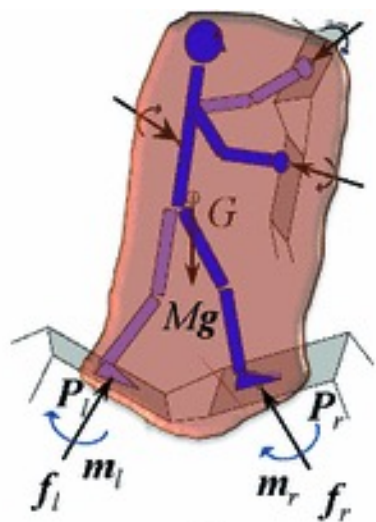


Cassie

Simplified dynamic models

□ Centroid models

- Boston dynamics
- MIT
- ETH



Angular/linear
positions

$$\begin{cases} \dot{\theta} = T(\theta)\omega \\ \dot{p} = R_{WB}(\theta)v \end{cases}$$

Angular/linear
velocities

$$\begin{cases} \dot{\omega} = I_{\text{nom}}^{-1} \left(-\omega \times I_{\text{nom}} \omega + \sum_{i=1}^{n_e} r_{E_i}(q_j) \times \lambda_{E_i} \right) \\ \dot{v} = g(\theta) + \frac{1}{m} \sum_{i=1}^{n_e} \lambda_{E_i} \end{cases}$$

Planning algorithms

- Convex model predictive control (MPC)
 - Quadratic programming (QP)

$$\begin{aligned} u[0 : T], x[0 : T] = & \underset{u[0:T], x[0:T]}{\operatorname{argmin}} \sum_{k=0}^T \underbrace{x^T Q x + u^T R u}_{\text{Quadratic costs}} \\ \text{s.t. } & \underbrace{x[k+1] = Ax[k] + Bu[k]}_{\text{Linear dynamics}} \\ & \underbrace{x[0] = x_0}_{\text{Linear constraints}} \\ & \underbrace{Cx + Du \leq b}_{\text{Linear constraints}} \end{aligned}$$

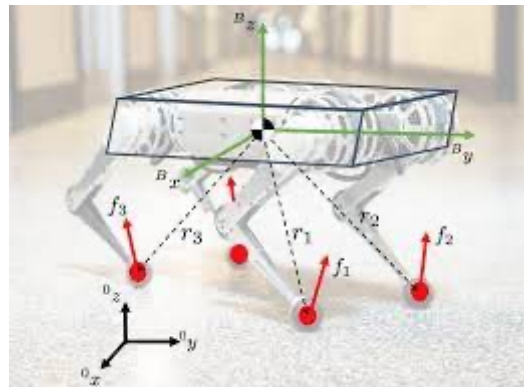
Planning algorithms

- Convex model predictive control (MPC)
 - Quadratic programming (QP)

$$\min_{\mathbf{x}, \mathbf{f}} \sum_{k=0}^m \|\mathbf{x}(k+1) - \mathbf{x}^{\text{ref}}(k+1)\|_{\mathbf{Q}} + \|\mathbf{f}(k)\|_{\mathbf{R}}$$

$$|f_x| \leq \mu f_z, \quad |f_y| \leq \mu f_z, \quad f_z > 0.$$

$$\mathbf{x}(k+1) = \mathbf{A}_k \mathbf{x}(k) + \mathbf{B}_k \hat{\mathbf{f}}(k) + \hat{\mathbf{g}}$$



Planning algorithms

□ Nonlinear MPC

- Solving the multi-step nonlinear trajectory optimization problem

$$\begin{array}{ll} \underset{\mathbf{u}(\cdot)}{\text{minimize}} & \Phi(\mathbf{x}(T)) + \int_0^T L(\mathbf{x}(t), \mathbf{u}(t), t) \, dt, \\ \text{subject to:} & \mathbf{x}(0) = \hat{\mathbf{x}}, \\ & \dot{\mathbf{x}} = \mathbf{f}^c(\mathbf{x}, \mathbf{u}, t), \\ & \mathbf{g}(\mathbf{x}, \mathbf{u}, t) = \mathbf{0}, \end{array}$$

Planning algorithms

□ Nonlinear MPC

■ Sequential quadratic programming (**SQP**)

$$\begin{array}{ll} \min_d & f(x_k) + \nabla f(x_k)^T d + \frac{1}{2} d^T \nabla^2 f(x_k) d & \text{Quadratize costs} \\ \text{s. t.} & b(x_k) + \nabla b(x_k)^T d \geq 0 & \text{Linearize model and constraints} \\ & c(x_k) + \nabla c(x_k)^T d = 0. & \\ \hline & \text{Iterates} & \end{array}$$

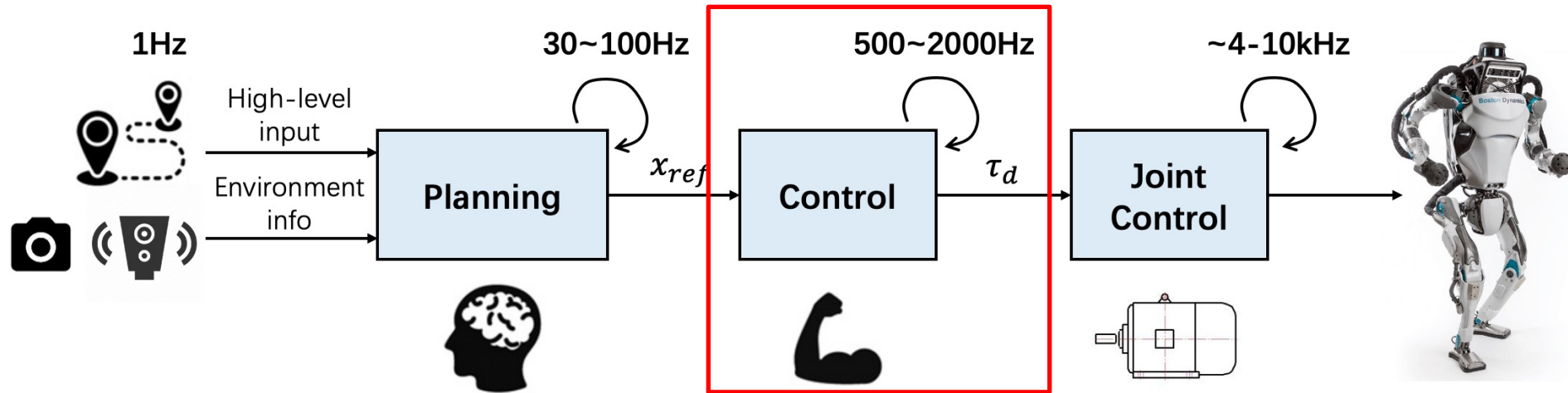
■ Differential dynamic programming (**DDP**)

- Basically iterative LQR (**ILQR**)

$$P = A^T P A + Q - A^T P B [B^T P B + R]^{-1} B^T P A$$

Tracking control

□ Grounding planned action to torque command



$$u(\cdot), x(\cdot) = \underset{u(\cdot), x(\cdot)}{\operatorname{argmin}} l_T(x(T)) + \int_{t=0}^T l(x(t), u(t))$$

$$\text{s.t.} \begin{cases} M(q)\ddot{q} + C(q, \dot{q})\dot{q} = \tau_g(q) + Bu + J^T(q)\tau_c \\ \phi(q) \geq 0 \quad \phi(q)\tau_c = 0 \\ \tau_c \in \mathcal{C}(n, \mu) \end{cases}$$

$$u \rightarrow \tau_d$$

Tracking control

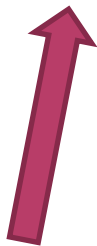
□ Inverse dynamics control

$$\begin{aligned}
 D(q)\ddot{q} + H(q, \dot{q}) &= Bu + J^T \lambda \\
 J(q)\ddot{q} + \dot{J}(q)\dot{q} &= 0
 \end{aligned}
 \quad
 \begin{bmatrix} \tilde{D}_{11} & \tilde{D}_{12} \\ \tilde{D}_{21} & \tilde{D}_{22} \end{bmatrix}
 \begin{bmatrix} \ddot{q}_c \\ \lambda \end{bmatrix}
 + \begin{bmatrix} \tilde{H}_1 \\ \tilde{H}_2 \end{bmatrix}
 = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} u$$



$$\begin{bmatrix} D & -J^T \\ J & 0 \end{bmatrix}
 \begin{bmatrix} \ddot{q} \\ \tau_F \end{bmatrix}
 + \begin{bmatrix} H(q, \dot{q}) \\ \dot{J}(q)\dot{q} \end{bmatrix}
 = \begin{bmatrix} B \\ 0 \end{bmatrix} u$$

Actuated joint torques





$$u = \bar{B}^{-1}(\bar{D} \ddot{q}_c + \bar{H})$$

$$\bar{D} = \tilde{D}_{11} - \tilde{D}_{12}\tilde{D}_{22}^{-1}\tilde{D}_{21}$$

$$\bar{H} = \tilde{H}_1 - \tilde{D}_{12}\tilde{D}_{22}^{-1}\tilde{H}_2$$

$$\bar{B} = \tilde{H}_1 - \tilde{D}_{12}\tilde{D}_{22}^{-1}\tilde{B}_2$$

Task space objectives



$$\begin{bmatrix}
 \text{torso pitch} \\
 \text{torso roll} \\
 \text{stance hip yaw} \\
 \text{swing hip yaw} \\
 p_{\text{st} \rightarrow \text{CoM}}^z \\
 p_{\text{sw} \rightarrow \text{CoM}}^x \\
 p_{\text{sw} \rightarrow \text{CoM}}^y \\
 p_{\text{sw} \rightarrow \text{CoM}}^z \\
 \text{swing toe absolute pitch}
 \end{bmatrix}$$

Tracking control

□ Whole Body Control (WBC)

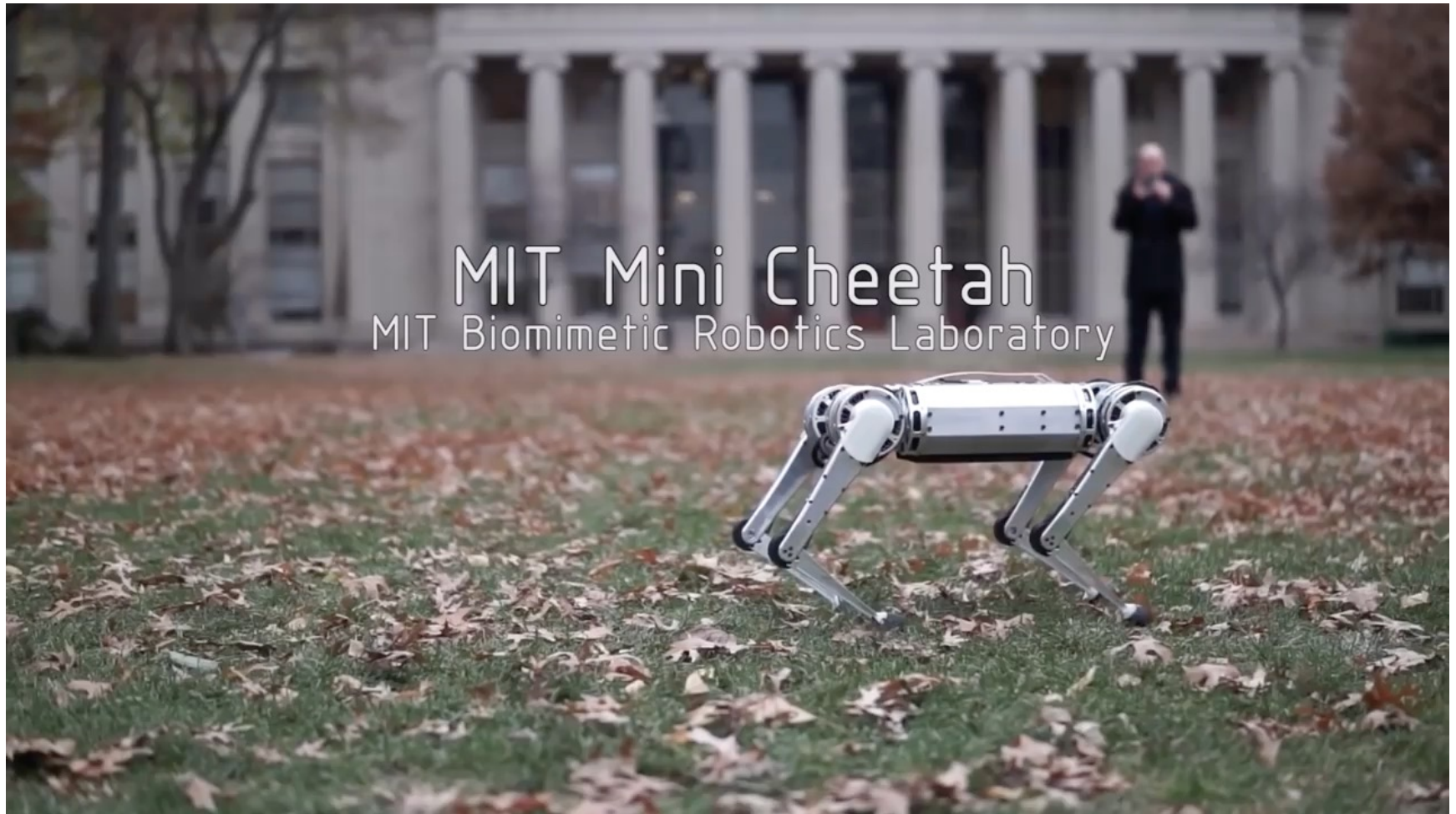
- Track the planned trajectory at the first time step using full dynamics
- Consider physical constraints

$$\begin{aligned} \underset{\ddot{\mathbf{q}}, \boldsymbol{\tau}, \mathbf{F}_c}{\text{minimize}} \quad & \sum_i \left\| \mathbf{J}_i \ddot{\mathbf{q}} + \dot{\mathbf{J}}_i \dot{\mathbf{q}} - \dot{\mathbf{v}}_i^{\text{cmd}} \right\|_{Q_i} \\ & + \left\| \mathbf{A} \ddot{\mathbf{q}} + \dot{\mathbf{A}} \dot{\mathbf{q}} - \dot{\mathbf{h}}^{\text{cmd}} \right\|_{Q_h} \\ & + \left\| \mathbf{F}_c - \mathbf{F}_c^{\text{cmd}} \right\|_{R_F} + \left\| \boldsymbol{\tau} \right\|_{R_\tau} \\ \text{subject to} \quad & \text{Robot Full Dynamics (10)} \\ & \mathbf{J}_c \ddot{\mathbf{q}} + \dot{\mathbf{J}}_c \dot{\mathbf{q}} = \mathbf{0} \\ & \mathbf{F}_c \in \mathcal{F} \\ & \left\| \boldsymbol{\tau} \right\| \leq \tau_{\max} \end{aligned}$$

$$\boldsymbol{\tau} = \mathbf{M}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + \mathbf{c}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) + \mathbf{g}(\boldsymbol{\theta}) + \mathbf{J}^T(\boldsymbol{\theta})\mathcal{F}_{\text{tip}}$$

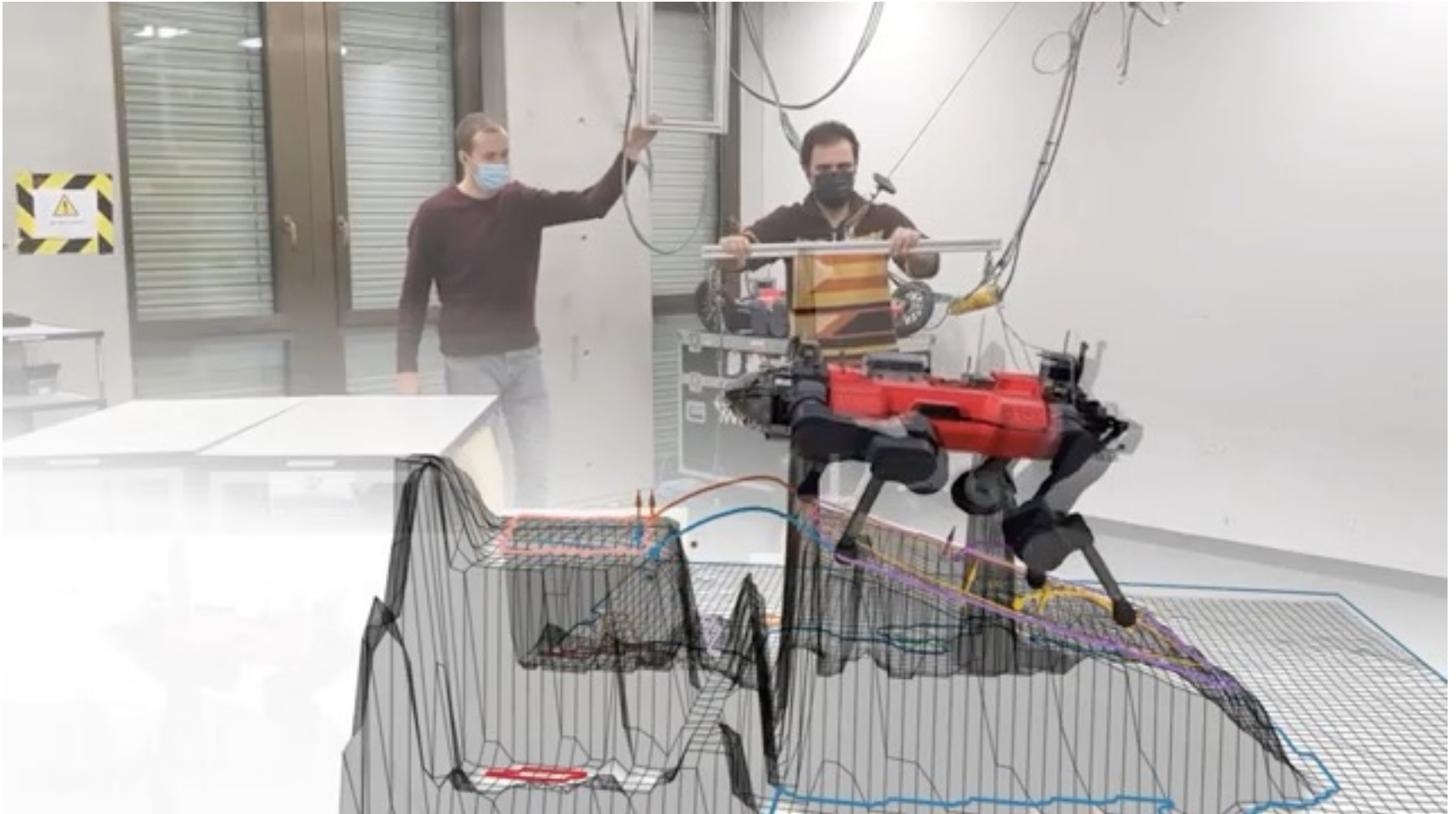
Examples

□ MIT Mini-cheetah (SRB+CMPC+ID/WBC)



Example

□ ETH Anymal (Centroid+NMPC+WBC)



Example

□ BD Atlas (Offline planning+NMPC+WBC)

Including Kinematics in MPC

Boston Dynamics

Centroidal MPC + IK

Centroidal + Kinematic MPC

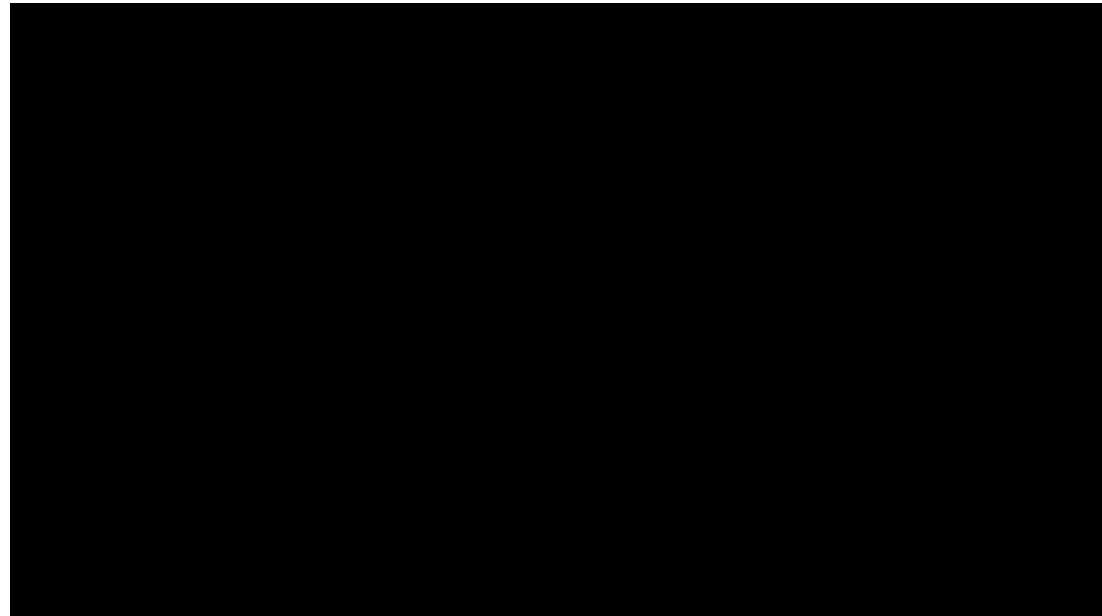
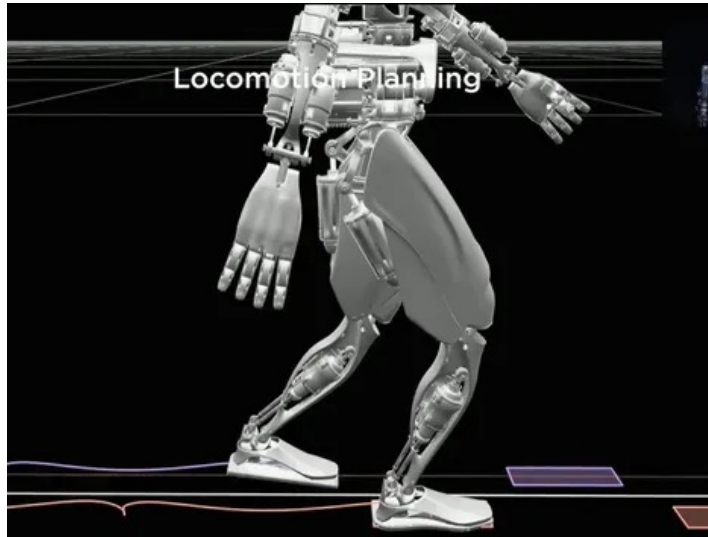
The image compares two motion planning approaches for a humanoid robot. The left side, titled 'Centroidal MPC + IK', shows a sequence of two frames where a grey robot model reaches for a green vertical bar. In the second frame, a red semi-transparent overlay shows the robot's body in a different pose, indicating a kinematic correction. Below this is a video frame showing a real robot in a similar pose. The right side, titled 'Centroidal + Kinematic MPC', shows the same grey robot model reaching for the green bar, but with a plus sign and a separate image of a red and blue semi-transparent overlay showing a more complex, multi-pose kinematic correction. Below this is a black video frame. The Boston Dynamics logo is in the top right. The bottom left corner has the text 'ICRA 2022: Atlas' and the bottom right corner has '2022-05-27 / SLIDE 19'.

ICRA 2022: Atlas

2022-05-27 / SLIDE 19

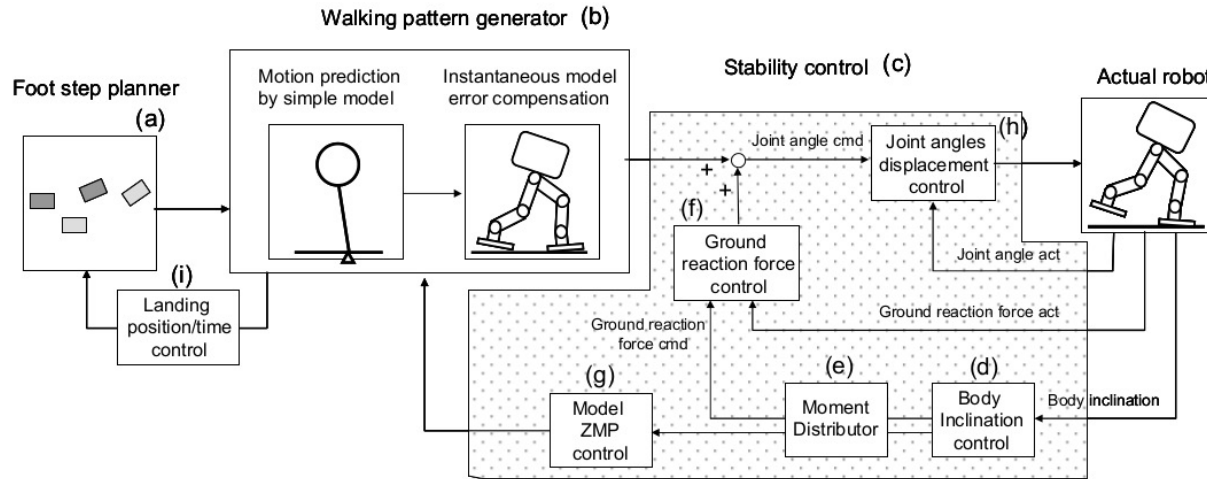
Example

□ Tesla Optimus (Offline planning+position control)



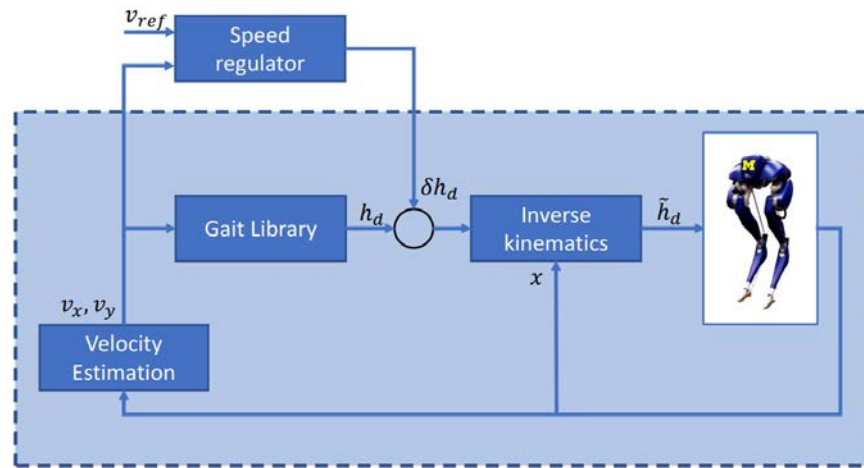
Example

□ Honda ASIMO (Simplified model+position control)



Example

□ Agility Robotics (Offline gait+Inverse dynamics control)



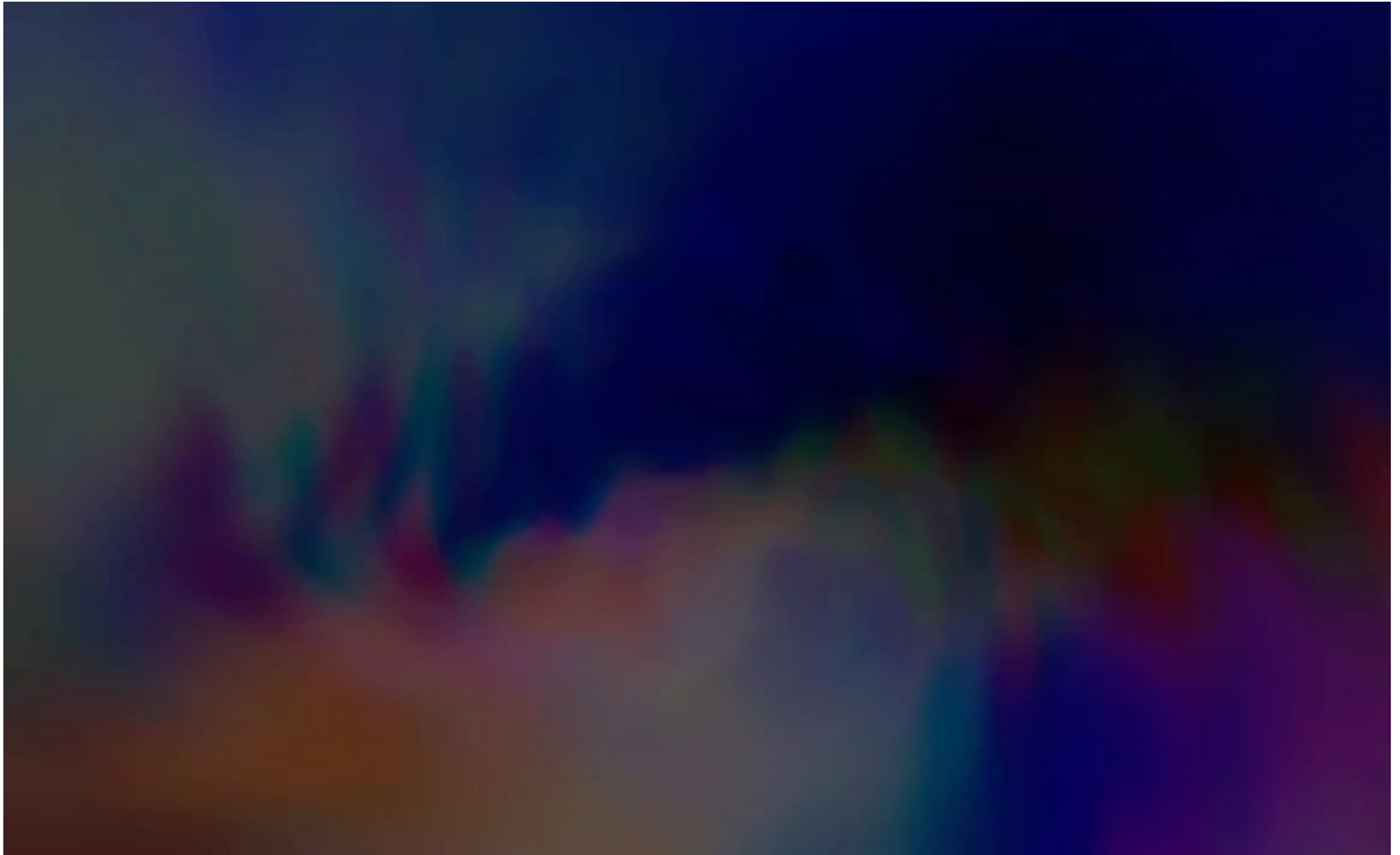
Example

□ Xiaoxing V1 (Simplified model+ID)



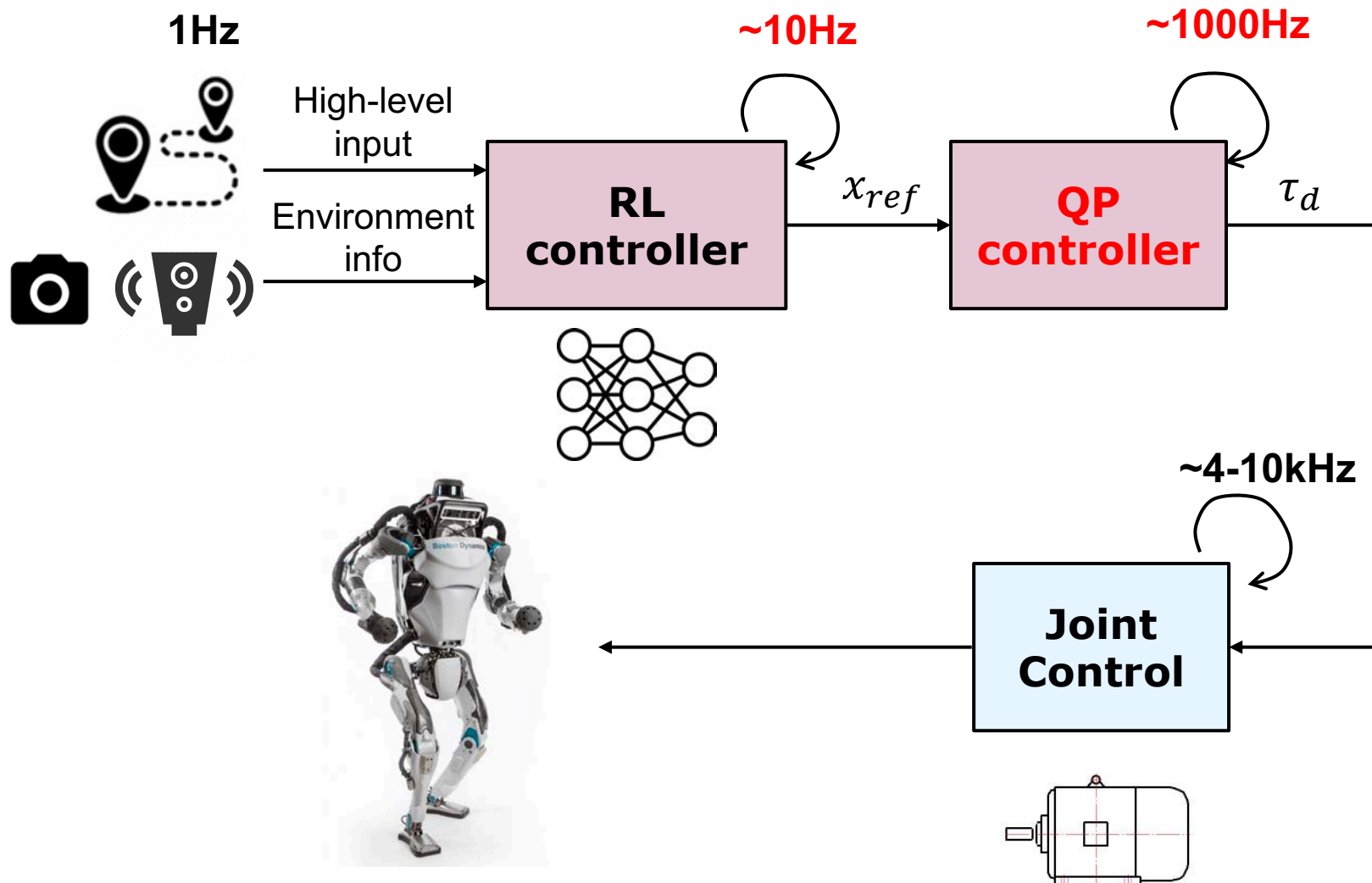
Example

□ Xiaoxing V2 (MPC+WBC)



RL with QP controller

- Utilize the robot dynamics model with RL



Example

□ RL with QP controller



Conference on
Robot Learning

amazonrobotics

Aurora

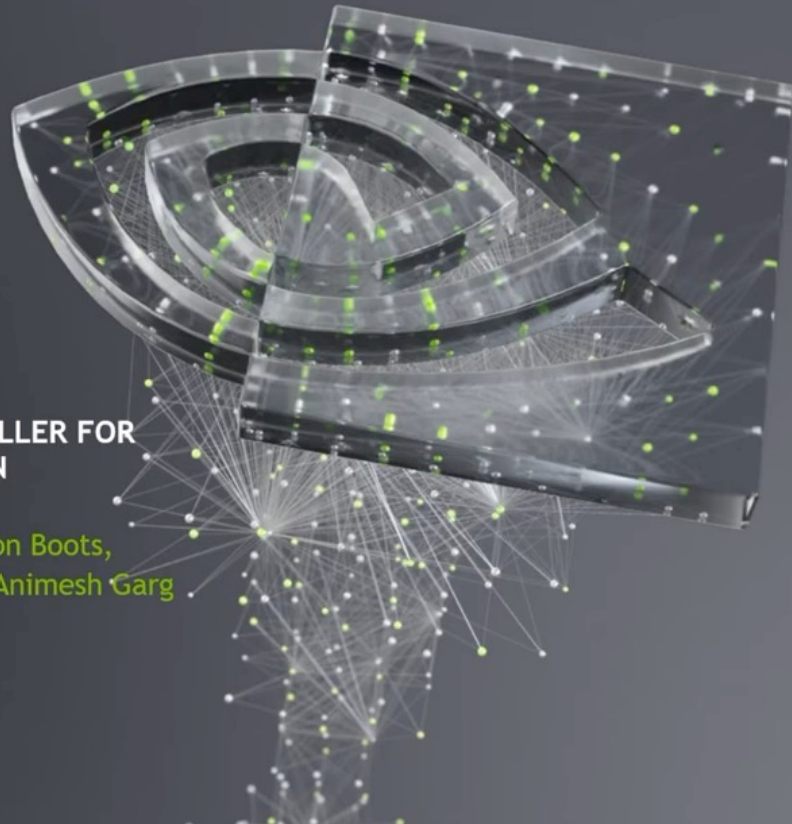


NVIDIA

LEARNING A CONTACT-ADAPTIVE CONTROLLER FOR
ROBUST, EFFICIENT LEGGED LOCOMOTION

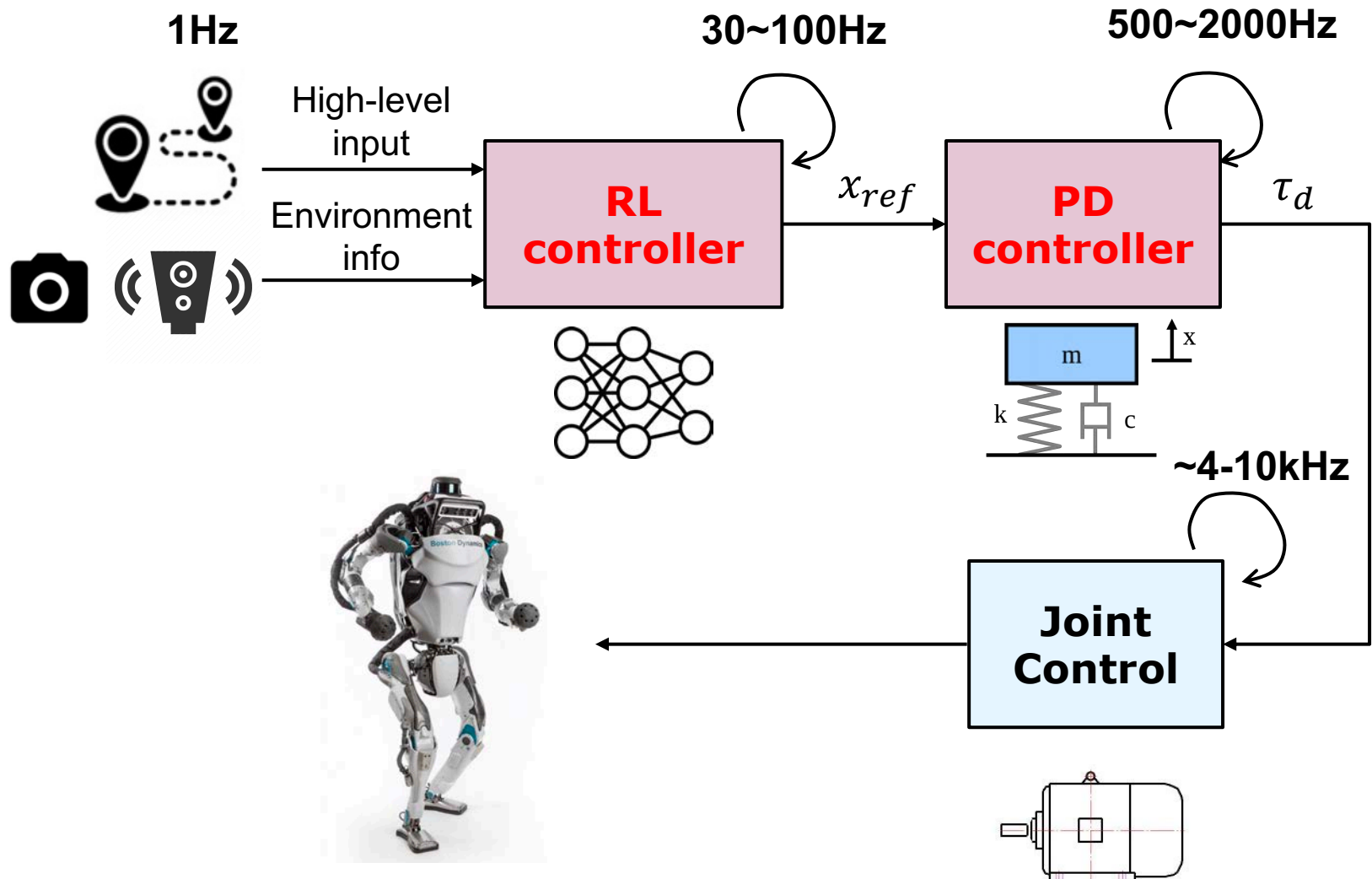
Xingye Da, Zhaoming Xie, David Hoeller, Byron Boots,
Anima Anandkumar, Yuke Zhu, Buck Babich, Animesh Garg

CoRL 2020



RL with low-level controller

- The most commonly used framework with RL



ARTIFICIAL INTELLIGENCE

Learning agile and dynamic motor skills for legged robots

Jemin Hwangbo^{1*}, Joonho Lee¹, Alexey Dosovitskiy², Dario Bellicoso¹, Vassilios Tsounis¹, Vladlen Koltun³, Marco Hutter¹

Legged robots pose one of the greatest challenges in robotics. Dynamic and agile maneuvers of animals cannot be imitated by existing methods that are crafted by humans. A compelling alternative is reinforcement learning, which requires minimal craftsmanship and promotes the natural evolution of a control policy. However, so far, reinforcement learning research for legged robots is mainly limited to simulation, and only few and comparably simple examples have been deployed on real systems. The primary reason is that training with real robots, particularly with dynamically balancing systems, is complicated and expensive. In the present work, we introduce a method for training a neural network policy in simulation and transferring it to a state-of-the-art legged system, thereby leveraging fast, automated, and cost-effective data generation schemes. The approach is applied to the ANYmal robot, a sophisticated medium-dog-sized quadrupedal system. Using policies trained in simulation, the quadrupedal machine achieves locomotion skills that go beyond what had been achieved with prior methods: ANYmal is capable of precisely and energy-efficiently following high-level body velocity commands, running faster than before, and recovering from falling even in complex configurations.

Basic walking

□ Learning agile and dynamic motor skills for legged robots

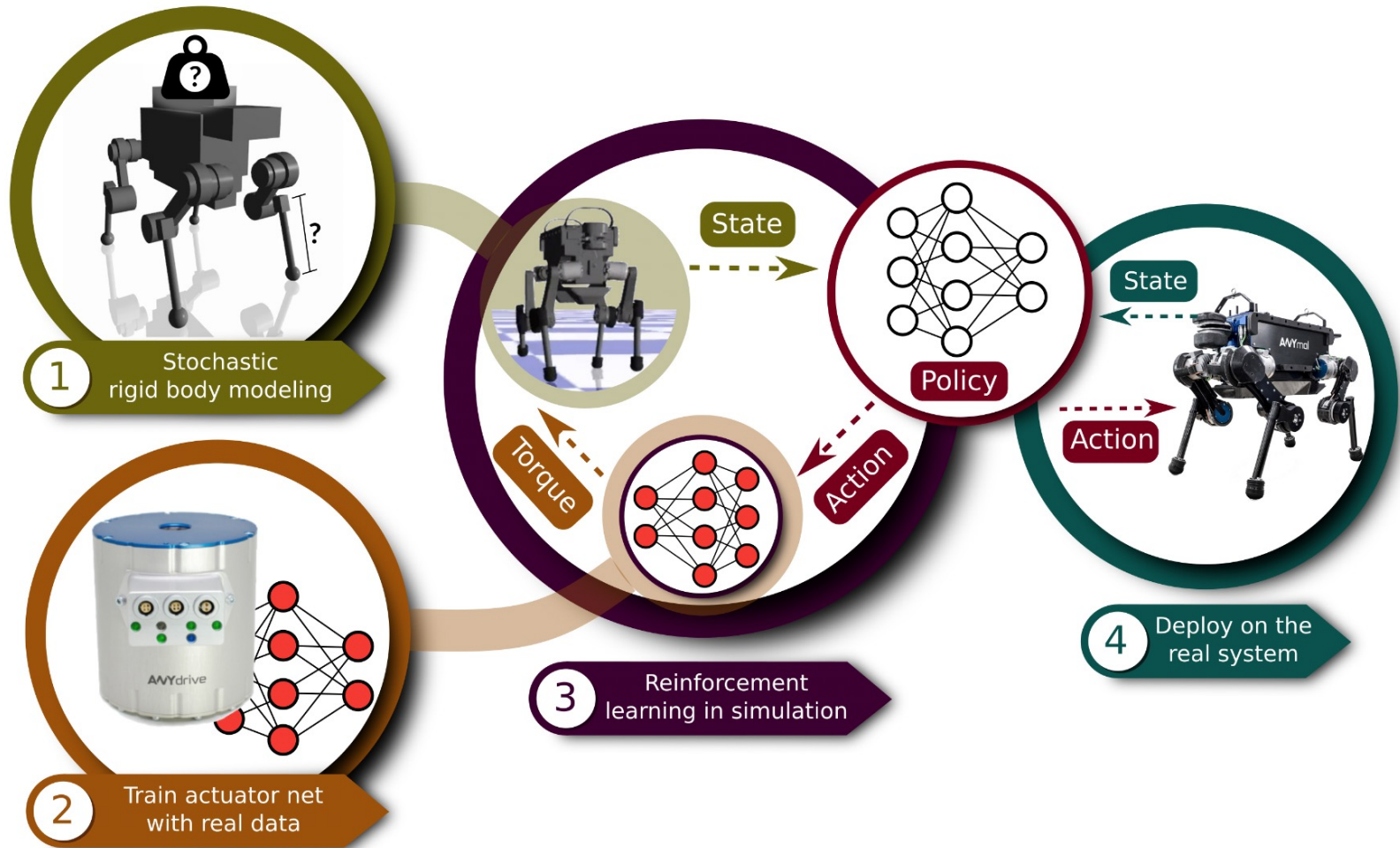


Fig. 1. Creating a control policy. In the first step, we identify the physical parameters of the robot and estimate uncertainties in the identification. In the second step, we train an actuator net that models complex actuator/software dynamics. In the third step, we train a control policy using the models produced in the first two steps. In the fourth step, we deploy the trained policy directly on the physical system.

Basic walking

Learning to Walk in Minutes Using Massively Parallel Deep Reinforcement Learning

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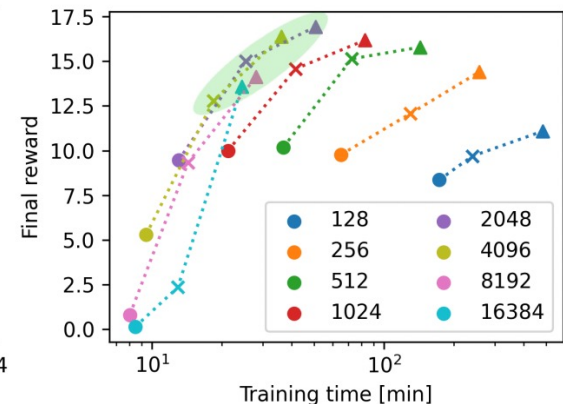
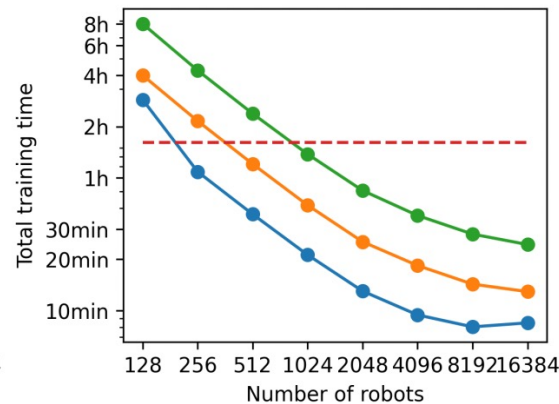
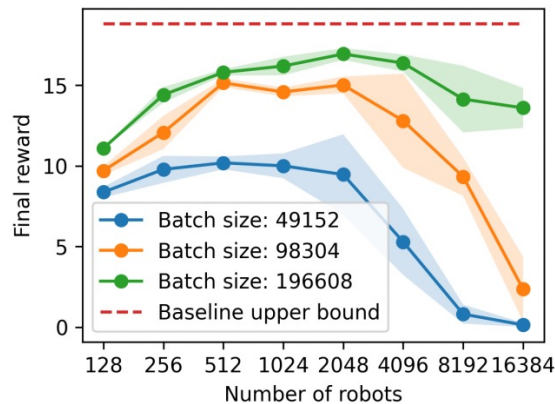
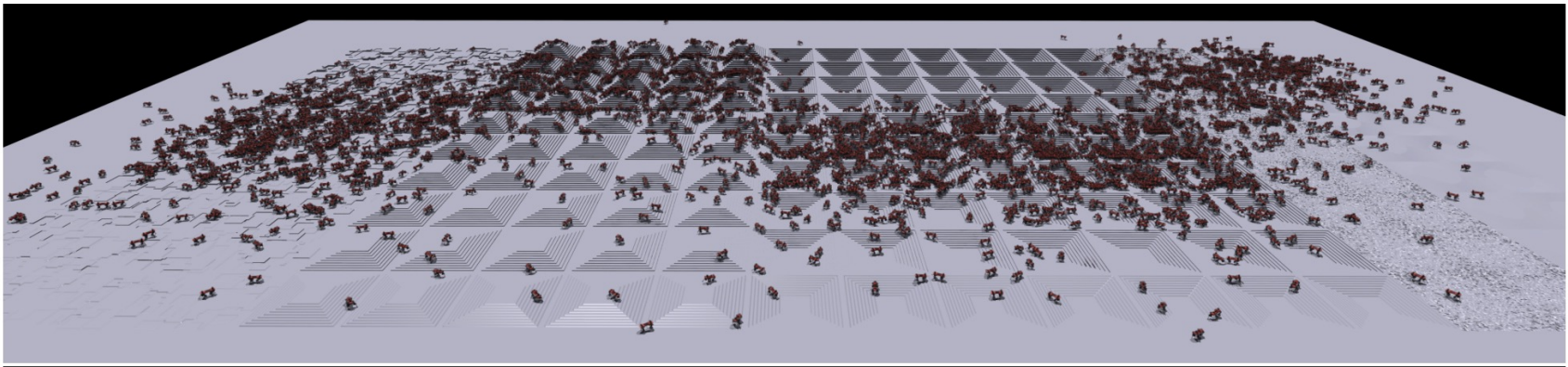
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Abstract: In this work, we present and study a training set-up that achieves fast policy generation for real-world robotic tasks by using massive parallelism on a single workstation GPU. We analyze and discuss the impact of different training algorithm components in the massively parallel regime on the final policy performance and training times. In addition, we present a novel game-inspired curriculum that is well suited for training with thousands of simulated robots in parallel. We evaluate the approach by training the quadrupedal robot ANYmal to walk on challenging terrain. The parallel approach allows training policies for flat terrain in under four minutes, and in twenty minutes for uneven terrain. This represents a speedup of multiple orders of magnitude compared to previous work. Finally, we transfer the policies to the real robot to validate the approach. We open-source our training code to help accelerate further research in the field of learned legged locomotion: https://leggedrobotics.github.io/legged_gym/.

Basic walking

□ Learning to Walk in Minutes Using Massively Parallel Deep Reinforcement Learning



Advancing Humanoid Locomotion: Mastering Challenging Terrains with Denoising World Model Learning

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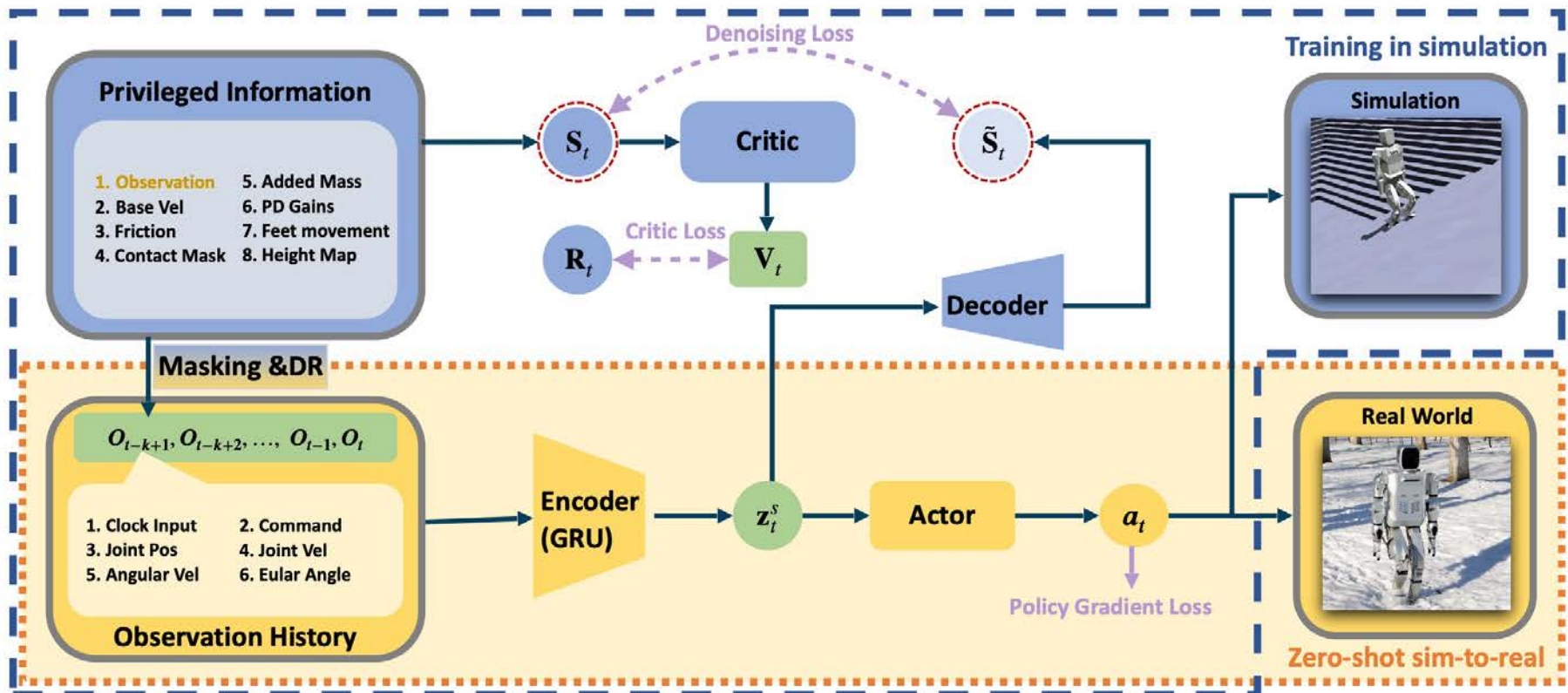
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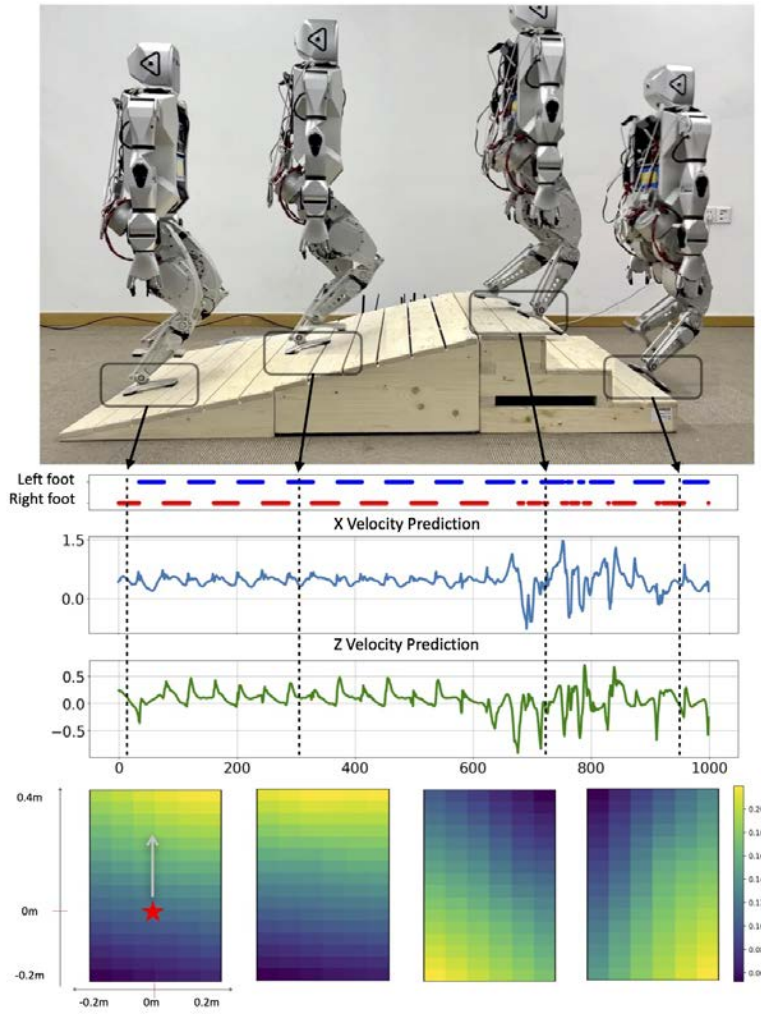
[†] Corresponding Author.

Challenging terrain

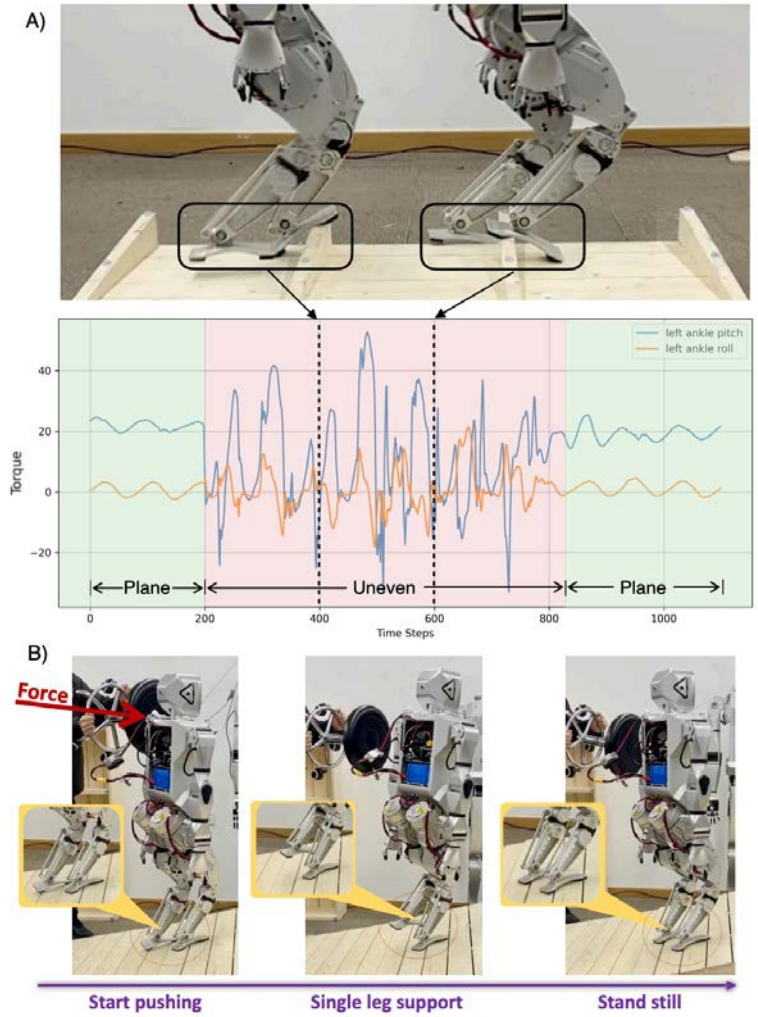
Advancing Humanoid Locomotion: Mastering Challenging Terrains with Denoising World Model Learning



Challenging terrain



Environment information recovery



Adaptive ankle control

ANIMAL ROBOTS

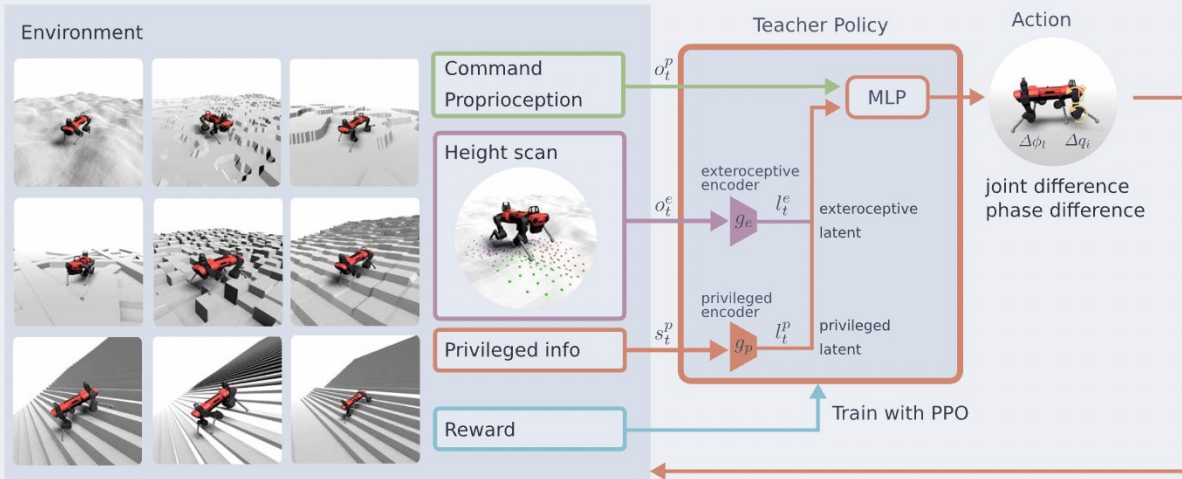
Learning robust perceptive locomotion for quadrupedal robots in the wild

Takahiro Miki^{1*}, Joonho Lee¹, Jemin Hwangbo², Lorenz Wellhausen¹, Vladlen Koltun³, Marco Hutter¹

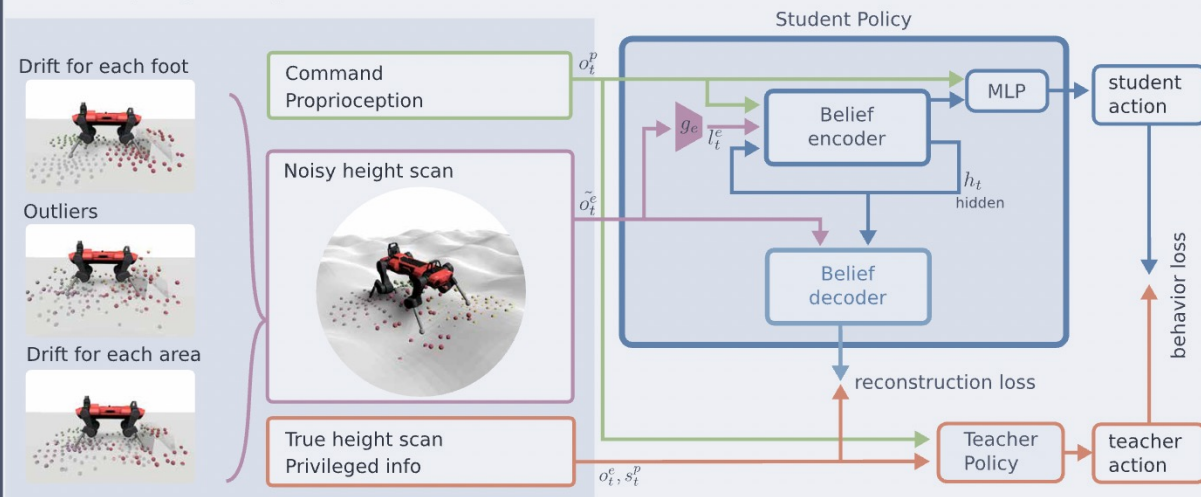
Legged robots that can operate autonomously in remote and hazardous environments will greatly increase opportunities for exploration into underexplored areas. Exteroceptive perception is crucial for fast and energy-efficient locomotion: Perceiving the terrain before making contact with it enables planning and adaptation of the gait ahead of time to maintain speed and stability. However, using exteroceptive perception robustly for locomotion has remained a grand challenge in robotics. Snow, vegetation, and water visually appear as obstacles on which the robot cannot step or are missing altogether due to high reflectance. In addition, depth perception can degrade due to difficult lighting, dust, fog, reflective or transparent surfaces, sensor occlusion, and more. For this reason, the most robust and general solutions to legged locomotion to date rely solely on proprioception. This severely limits locomotion speed because the robot has to physically feel out the terrain before adapting its gait accordingly. Here, we present a robust and general solution to integrating exteroceptive and proprioceptive perception for legged locomotion. We leverage an attention-based recurrent encoder that integrates proprioceptive and exteroceptive input. The encoder is trained end to end and learns to seamlessly combine the different perception modalities without resorting to heuristics. The result is a legged locomotion controller with high robustness and speed. The controller was tested in a variety of challenging natural and urban environments over multiple seasons and completed an hour-long hike in the Alps in the time recommended for human hikers.

Perceptive locomotion

1. Teacher policy training



2. Student policy training



Whole body control

Combining Learning-based Locomotion Policy with Model-based Manipulation for Legged Mobile Manipulators

Yuntao Ma, Farbod Farshidian, Takahiro Miki, Joonho Lee, Marco Hutter

Abstract—Deep reinforcement learning produces robust locomotion policies for legged robots over challenging terrains. To date, few studies have leveraged model-based methods to combine these locomotion skills with the precise control of manipulators. Here, we incorporate external dynamics plans into learning-based locomotion policies for mobile manipulation. We train the base policy by applying a random wrench sequence on the robot base in simulation and add the noisified wrench sequence prediction to the policy observations. The policy then learns to counteract the partially-known future disturbance. The random wrench sequences are replaced with the wrench prediction generated with the dynamics plans from model predictive control to enable deployment. We show zero-shot adaptation for manipulators unseen during training. On the hardware, we demonstrate stable locomotion of legged robots with the prediction of the external wrench.

I. INTRODUCTION

Legged robots have potential to traverse rough unstructured terrain that are unreachable by traditional wheeled or tracked robots, utilize human-oriented infrastructures such as stairs and doors, and conduct various tasks by carrying task-specific payloads in harsh environments instead of human.

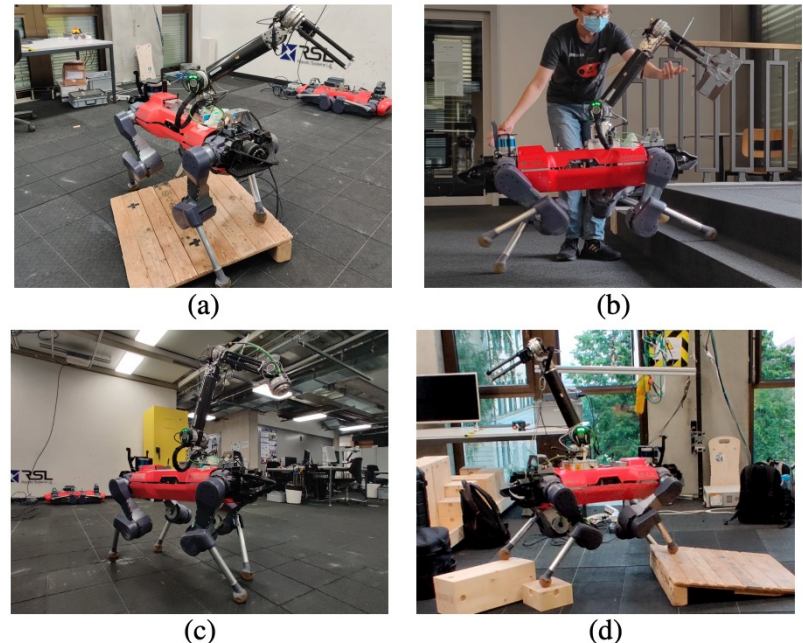
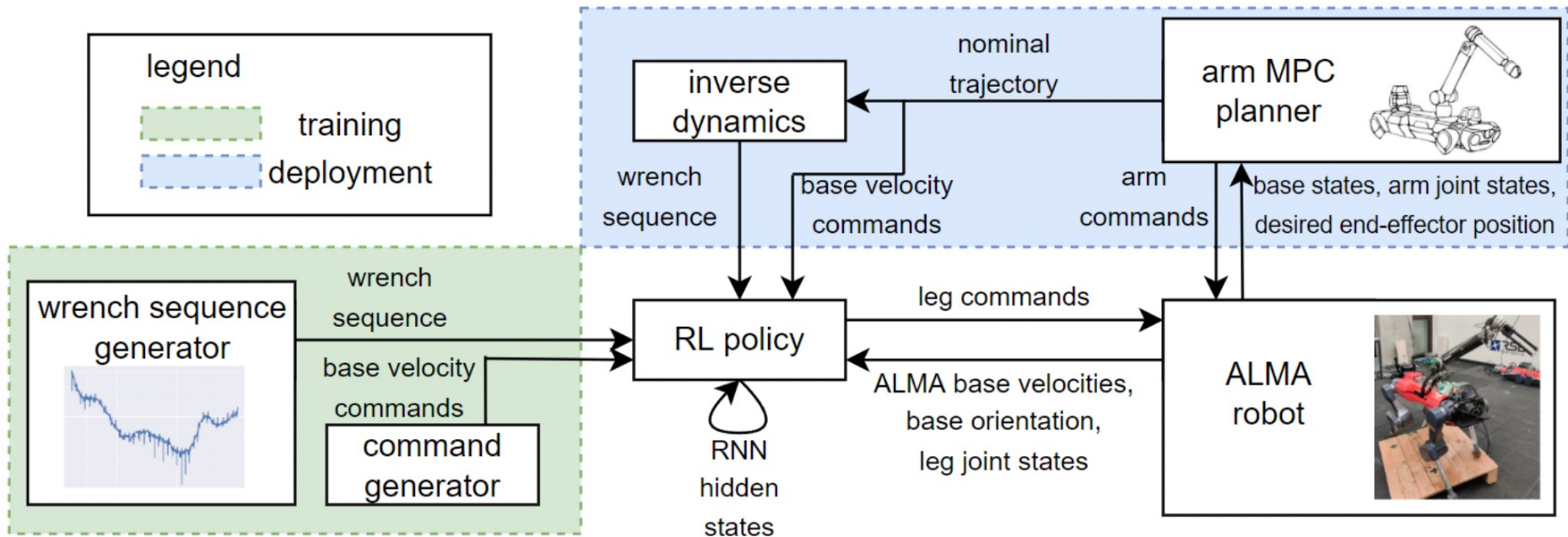


Fig. 1: Experiments with ALMA. (1a-1c) ALMA trotting with different manipulator configurations: no end-effector, fixed brick, a wrist weighing 1.8 kg. (1d) ALMA trotting on challenging terrain.

Whole body control

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Whole body control

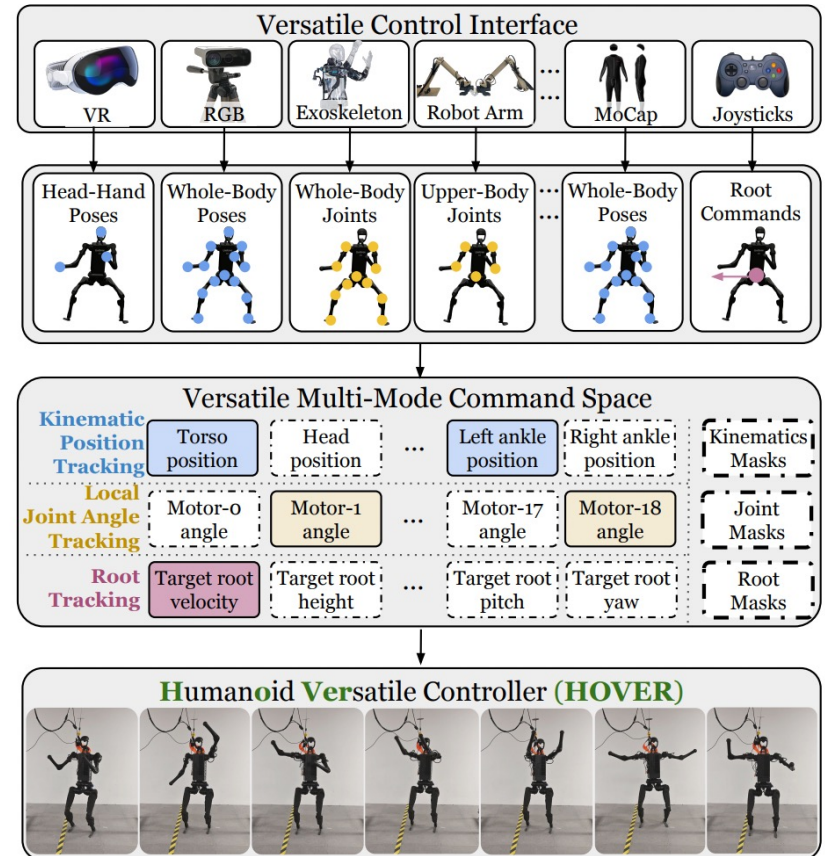
HOVER: Versatile Neural Whole-Body Controller for Humanoid Robots

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Jan Kautz¹ Changliu Liu² Guanya Shi² Xiaolong Wang^{1,5} Linxi “Jim” Fan^{†1} Yuke Zhu^{†1,4}

Abstract—Humanoid whole-body control requires adapting to diverse tasks such as navigation, loco-manipulation, and tabletop manipulation, each demanding a different mode of control. For example, navigation relies on root velocity or position tracking, while tabletop manipulation prioritizes upper-body joint angle tracking. Existing approaches typically train individual policies tailored to a specific command space, limiting their transferability across modes. We present the key insight that full-body kinematic motion imitation can serve as a common abstraction for all these tasks and provide general-purpose motor skills for learning multiple modes of whole-body control. Building on this, we propose HOVER (Humanoid Versatile Controller), a multi-mode policy distillation framework that consolidates diverse control modes into a unified policy. HOVER enables seamless transitions between control modes while preserving the distinct advantages of each, offering a robust and scalable solution for humanoid control across a wide range of modes. By eliminating the need for policy retraining for each control mode, our approach improves efficiency and flexibility for future humanoid applications.

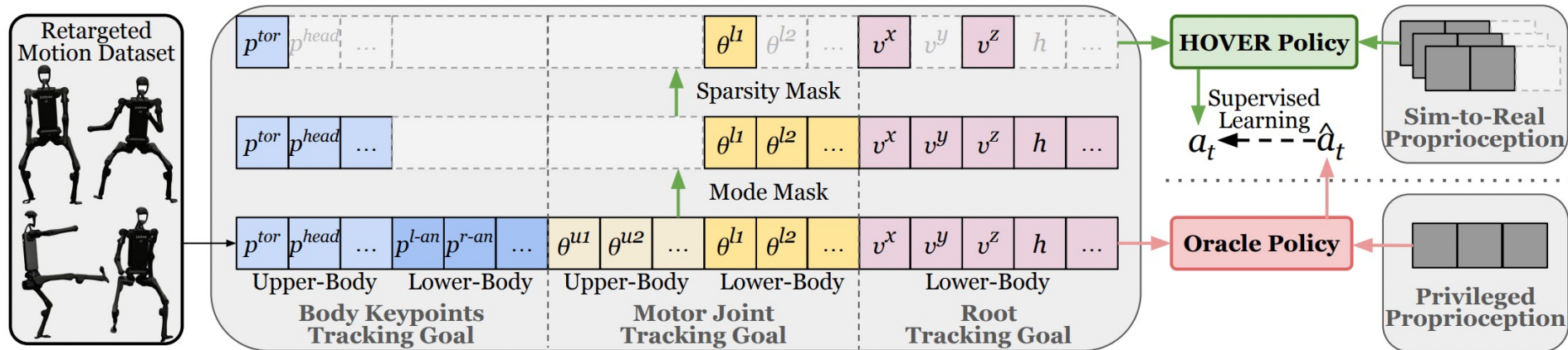
I. INTRODUCTION

Humanoid is a versatile form factor that supports a wide variety of robotic tasks and applications, including bimanual



Whole body control

- Hover: versatile neural whole-body controller for humanoid robots



Thank you!



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